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Family-wise error control for correlated pathway tests in GWAS: CATFISH with Westfall–Young calibration

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Abstract

Motivation: Pathway analysis of genome-wide association study (GWAS) summary statistics must contend with two distinct sources of dependence. Genes within a pathway are correlated through linkage disequilibrium (LD), and pathways are correlated with one another through shared gene membership. Most pipelines address only the first, calibrating each pathway against its own LD-aware null and then applying a Bonferroni or Benjamini–Hochberg correction that assumes the resulting pathway p-values are independent. They are not, and the correction is consequently miscalibrated in a direction that depends on how strongly the pathway family overlaps.

Results: We extend CATFISH, a correlation-aware omnibus pathway framework combining ACAT, Fisher, adaptive soft TFisher, minP and Stouffer statistics, with a Westfall–Young-style family-wise calibration. A single genome-wide correlated null draw, generated from a multivariate normal model of gene–gene correlation, is propagated through every pathway in the family; read down its columns it gives each pathway a calibrated marginal p-value, and read across its rows the minimum over pathways gives a family-wise threshold that accounts for between-pathway dependence without modelling it. Numerical precision is treated as a correctness requirement: in a GWAS of $n \approx 1.3$ million individuals the omnibus statistic underflows double precision for the majority of pathways, and the resulting ties silently destroy the ranking. Across simulated null families spanning three LD regimes and three pathway sizes, independence-based calibration was anti-conservative while per-pathway and Westfall–Young calibration held type I and family-wise error near nominal, and both remained stable when up to four of the five component tests were deliberately miscalibrated. Under five stylised signal architectures, the component predicted to be sensitive to each was the best-performing component in 15 of 15 architecture-by-size combinations. Applied to a maize grain-nitrogen GWAS, the Westfall–Young ordering recovered L-homoserine biosynthesis as the leading pathway and placed a known aspartate kinase isozyme first among 34,873 genes. Applied to a GLGC LDL-cholesterol GWAS, it exposed a regime in which the self-contained null is false almost everywhere: 11.2% of genes reach genome-wide significance, essentially every pathway is significant, and exact FWER control does not rescue the ranking. We argue this is a property of self-contained testing at very large n, not a failure of multiple-testing correction.

Availability: CATFISH is an R package available at `https://github.com/nirwantandukar/CATFISH` under the MIT licence. All simulation, analysis and figure code for this paper is in the `catfish2/` directory of that repository.

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Supplementary information: Supplementary data are available at Bioinformatics online.

Keywords genome-wide association study, pathway analysis, gene-set test, family-wise error rate, Westfall–Young, permutation, ACAT, MAGMA

Abbreviations GWAS, MAGMA, LD, FWER, WY, MVN, ACAT

Key Messages

- Pathway p-values are correlated because pathways share genes; Bonferroni and Benjamini–Hochberg corrections applied to them are therefore miscalibrated, and the direction of the error depends on the overlap structure of the particular pathway database used.
- A Westfall–Young max-statistic null, built by propagating one genome-wide correlated draw through the whole pathway family, controls the family-wise error rate in the strong sense without assuming the pathways are independent; its validity rests instead on the multivariate normal correlation model used to generate the draw.
- Implementing the omnibus in log space is a correctness requirement, not an optimisation: at GWAS sample sizes above roughly one million the analytic omnibus underflows double precision and the pathway ranking collapses into ties.
- Self-contained pathway tests saturate at very large n. When 11% of all genes are genome-wide significant, a random 50-gene set contains several of them, almost every pathway rejects, and exact FWER control does not restore discrimination. Saturation is a property of the hypothesis being tested, not of the correction applied to it.

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1 Introduction

Genome-wide association studies (GWAS) have transformed complex trait research by identifying thousands of loci linked to disease, development, and quantitative phenotypes. Yet most variants explain only a small fraction of phenotypic variance and often fall in noncoding regions, limiting single-marker mechanistic interpretation (Manolio et al., 2009; Boyle et al., 2017). Complex traits are often highly polygenic, with relevant signals distributed across many modest-effect loci rather than concentrated in a few genome-wide significant variants (Akula et al., 2025; Lappalainen et al., 2024). Gene-set analysis, including pathway-based approaches, is therefore a key post-GWAS strategy for translating summary statistics into biological hypotheses. By aggregating evidence across predefined groups of functionally related genes, pathway-based methods enhance interpretability, reduce the multiple-testing burden, and reveal coordinated pathway disruptions that single-locus analyses may miss (Mooney and Wilmot, 2015; de Leeuw et al., 2015). This is especially valuable for polygenic traits, where many genes may influence the same phenotype through shared biological processes. In such settings, a candidate gene becomes more compelling when locus- or gene-level association evidence is reinforced by pathway-level enrichment from a biologically relevant process. Pathway support therefore provides functional context for prioritization, although it is best interpreted as complementary rather than definitive evidence because pathway annotations remain incomplete and often overlapping.

A plethora of pathway analysis tools has been developed for GWAS. Classical over-representation methods test whether predefined pathways are enriched among significant genes, as implemented in DAVID, g:Profiler, and WebGestalt (Sherman et al., 2022; Kolberg et al., 2023; Liao et al., 2019). Ranking-based methods such as Gene Set Enrichment Analysis (GSEA) assess whether pathway genes are non-randomly concentrated at the top or bottom of a ranked gene list (Subramanian et al., 2005). GWAS-focused extensions, including MAGENTA, INRICH, and MAGMA, add features such as SNP-to-gene mapping, pathway-level modeling, and adjustment for linkage disequilibrium (LD) or gene size (Segrè et al., 2010; Lee et al., 2012; de Leeuw et al., 2015). Accounting for LD is essential in GWAS pathway analysis because nearby variants are often correlated and therefore do not provide independent evidence of association. If LD is ignored, clusters of linked SNPs can be effectively counted multiple times, leading to inflated gene-level aggregation and enrichment biases toward larger genes or regions with dense marker coverage. Several widely used post-GWAS methods therefore explicitly account for LD through model-based or reference-based dependence estimates rather than assuming independence. VEGAS uses multivariate normal simulation, Pascal uses LD from a reference population to compute gene and pathway scores, and MAGMA models gene-level dependence through an approximated correlation structure (Liu et al., 2010; de Leeuw et al., 2015; Lamparter et al., 2016; Watanabe et al., 2017).

These methods address dependence *within* a pathway. A second source of dependence operates *between* pathways and is far less often handled. Curated pathway databases are built from overlapping annotations: a gene that participates in several metabolic steps is a member of several pathways, and hierarchical resources such as Reactome place parent and child pathways in the same family. The pathway-level p-values returned by any method are therefore correlated with one another, sometimes strongly. Nonetheless, the final multiple-testing step in essentially every published pipeline is a Bonferroni or Benjamini–Hochberg adjustment of those p-values, both of which are derived under independence or under positive-regression dependence conditions that overlapping gene sets do not generally satisfy. The resulting family-wise error rate is not the nominal one, and the size of the discrepancy is a property of the particular database rather than of the method, which makes it invisible to the kind of simulation study that generates independent pathways.

The classical solution to exactly this problem is the Westfall–Young max-statistic procedure (Westfall and Young, 1993). Rather than correcting each p-value in isolation, one generates a null realization of the *entire* family of tests, records the single most extreme statistic observed in that realization, and repeats. The distribution of that maximum is the reference distribution for the observed minimum, and the resulting threshold controls the family-wise error rate under the complete null whatever the dependence structure among the tests, because the dependence is inherited from the way the null realizations are generated rather than assumed. In a GWAS pathway setting this means drawing correlated gene-level null statistics genome-wide, propagating that single draw through every pathway in the family, and recording the minimum pathway p-value. Nothing about the between-pathway correlation needs to be estimated, modelled, or even named.

Here we extend CATFISH (Combining **C**auchy combination test (ACAT), **A**daptive **T**Fisher (soft) test, **F**isher's test, m**I**n-P, and **S**touffer's method for **H**olistic pathway analysis), a correlation-aware pathway analysis framework that integrates multiple complementary gene-set statistics in a unified omnibus strategy, with Westfall–Young family-wise calibration. Different pathway tests are sensitive to different genetic architectures: some are best at detecting broad, distributed enrichment, whereas others are more sensitive to sparse strong signals or heterogeneous evidence. Instead of relying on a single summary statistic, CATFISH evaluates each pathway with several complementary tests and combines them in an omnibus framework. This improves robustness when the true signal configuration is unknown while preserving the interpretability of the component tests. Beyond pathway-level inference, CATFISH prioritizes candidate genes by integrating locus-level GWAS signals, gene-level MAGMA results, and pathway-level support.

This paper makes three contributions beyond the original CATFISH framework. First, we add the Westfall–Young layer described above and characterize how it behaves: the same correlated null, read down its columns, supplies each pathway's marginal p-value, and read across its rows, the family-wise one. Second, we show that the entire statistic must be computed in log space. This sounds like an implementation detail and is not: at the sample sizes now routine in human GWAS consortia, the analytic omnibus p-value for a typical pathway falls below the smallest representable double-precision number, every such pathway is assigned the same floor value, and the ranking that the whole analysis exists to produce is destroyed without any error being raised. Third, we report a negative result we believe is general. In a GLGC LDL-cholesterol GWAS of roughly 1.3 million individuals, the self-contained null hypothesis is false for almost every gene set that could be constructed, nearly every pathway is significant, and exact family-wise error control does not recover a usable ordering. We interpret this as saturation of the self-contained hypothesis rather than as a defect of the correction,

Table 1 Summary of pathway signal archetypes and expected CATFISH component sensitivity. Each archetype represents a conceptual ranked gene-level association profile rather than an empirically learned class of real pathways.

Archetype	Gene-level p-value pattern	Ranked-profile shape	Expected sensitive tests
Strong Sparse Signal (SSS)	A small number of genes have extremely small p-values, while most genes remain close to null.	Sharp early peak followed by a rapid drop; pronounced elbow.	ACAT, minP
Moderate Dense Signal (MDS)	Many genes show modest but consistent association, without one clearly dominant gene.	Broad shoulder of moderate signals across a substantial fraction of the pathway.	Fisher, Stouffer, soft TFisher
Weak Diffuse Enrichment (WDE)	Most genes are individually non-significant, but the pathway shows a subtle overall shift toward smaller p-values.	Gentle pathway-wide deflection with few or no strong individual peaks.	Stouffer, Fisher
Mixed Signal Architecture (MSA)	One or a few strong genes occur together with a broader set of moderate supporting signals.	Strong leading peak plus a moderate shoulder.	soft TFisher, Fisher, ACAT
Single-Gene Proxy (SGP)	One dominant gene accounts for most of the pathway-level signal, while the remaining genes are near null.	Single extreme peak followed by an otherwise null-like profile.	minP, ACAT

and discuss what it implies for the choice between self-contained and competitive testing as GWAS sample sizes continue to grow.

CATFISH is implemented as an R package and is freely available at `https://github.com/nirwantandukar/CATFISH` under the MIT licence.

1.1 Pathway signal archetypes

CATFISH was developed to capture heterogeneous pathway-level signal patterns, ranging from highly sparse gene-level associations to weak, broadly distributed enrichment. We define five conceptual pathway signal archetypes to clarify how different gene-level profiles are expected to favor different component tests. These archetypes are not intended as empirically learned classes of real pathways but as statistical templates for interpreting test behavior and motivating the complementary design of CATFISH.

For interpretive clarity, we consider a pathway-level working null and alternative. Under the null, gene-level evidence within a pathway matches the background distribution after accounting for dependence, with no systematic excess of small p-values. Under the alternative, the pathway shows enrichment through one or more forms of excess association evidence, including extreme individual genes, broad moderate signals, weak diffuse shifts, or mixed patterns. Figure 1 illustrates the five ranked signal shapes, and Table 1 summarizes their statistical profiles and expected CATFISH component sensitivity.

In short, Strong Sparse Signal (SSS) refers to pathways driven by a few strongly associated genes that favor extreme-tail methods. Single-Gene Proxy (SGP), a special case of SSS, is composed of a signal dominated by one leading gene, but it should be interpreted cautiously as gene-centric. Moderate Dense Signal (MDS) has many modest, consistent gene associations that favor aggregation methods. Weak Diffuse Enrichment (WDE) constitutes subtle pathway-wide shifts, with few individually significant genes, favoring distributed-signal methods. Mixed Signal Architecture (MSA) represents one or a few strong signals plus broader moderate support that favor balanced methods.

2 Methods

2.1 Gene-level inputs and component tests

For a pathway S containing $m = |S|$ genes, let p_g denote the gene-level association p-value for gene $g \in S$ and Z_g the corresponding gene-level Z-statistic. We write $\mathbf{p}_S = (p_1, \ldots, p_m)$ and $\mathbf{Z}_S = (Z_1, \ldots, Z_m)$. Gene-level statistics are obtained with MAGMA (de Leeuw et al., 2015), which aggregates SNP-level GWAS evidence across variants assigned to a gene while accounting for LD, gene size, and local marker density, and which additionally returns the gene–gene correlation matrix induced by LD. These quantities are the common inputs for every CATFISH component test and for both calibration layers.

Each pathway is assessed using five complementary component tests (Table 2): ACAT (Liu et al., 2019; Liu and Xie, 2020), Fisher (Fisher, 1932), adaptive soft-thresholding TFisher (Zhang et al., 2020), minP, and Stouffer (Stouffer et al., 1949). Together, they cover the archetypes in Section 1.1 and apply directly to gene-level p-values or Z-scores without external weights or pathway-specific tuning. We combine the component p-values into a single omnibus statistic via an ACAT-type cross-method operator $\mathcal{O}(\cdot)$. This omnibus is a pathway-level summary of evidence across components, not an additional gene-level test.

Four components (ACAT, Fisher, minP, Stouffer) are fully determined by gene-level p-values and require no tuning. Soft TFisher is the exception: it adapts over a grid of truncation thresholds τ, which determines which genes contribute. This is the only tunable choice in CATFISH, and we justify it below.

Following Zhang et al. (2020), the grid spans small, intermediate and large thresholds rather than concentrating at one end: we use $\tau \in \{10^{-2}, 5\times10^{-2}, 5\times10^{-1}\}$ and take the minimum calibrated TFisher p-value over the grid. The original authors' grid also includes $\tau = 1$, which we omit because $\tau = 1$ is Fisher's method and Fisher is already carried as a separate component. Spanning the range matters because the number of genes surviving truncation is Binomial(m, τ): a grid confined to small τ leaves $m\tau \ll 1$ at realistic set sizes, the statistic is then zero on a large fraction of pathways, and the component is degenerate precisely where it is most often applied.

A threshold is informative mainly when $m\tau$ is not far below one, so which τ values are active depends on set size: $\tau = 5 \times 10^{-1}$ is informative from the smallest sets upward, whereas $\tau = 10^{-2}$ requires $m \approx 100$. Adapting across τ rather than fixing a single value is what allows one component to serve both sparse and diffuse architectures, and users can assess the grid for their own database using $m\tau$.

Undefined components. When no gene in a pathway falls below any threshold in the grid, the soft TFisher statistic is zero at every τ and its one-sided p-value is exactly one. We treat this as *undefined* rather than as evidence against enrichment, and the omnibus for that pathway is formed from the components that are defined.

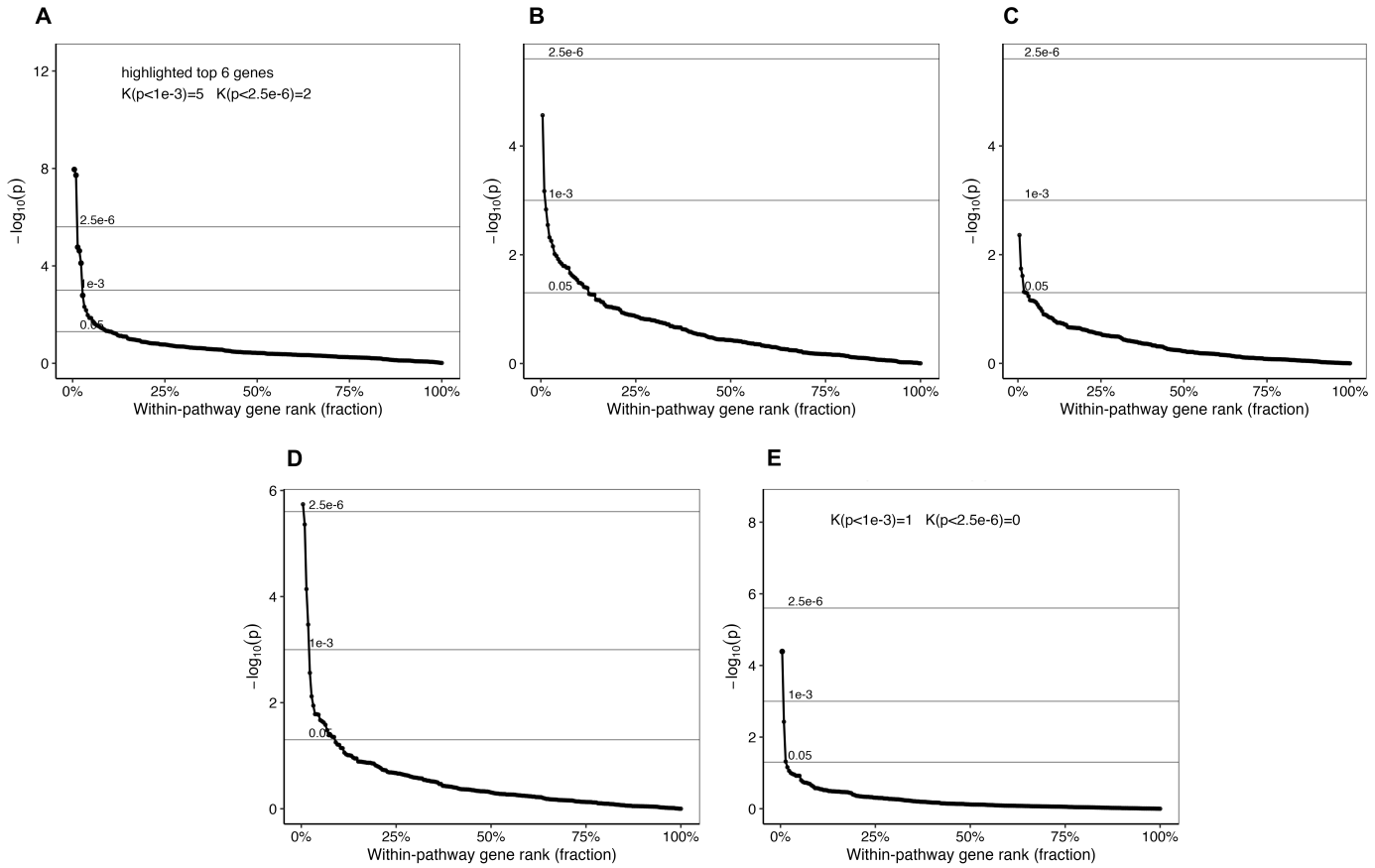


Figure 1 Ranked gene-level association profiles for five pathway signal archetypes. Panels illustrate conceptual within-pathway $-\log_{10}(p)$ profiles for Strong Sparse Signal (SSS), Moderate Dense Signal (MDS), Weak Diffuse Enrichment (WDE), Mixed Signal Architecture (MSA), and Single-Gene Proxy (SGP). Horizontal lines mark representative significance thresholds. These archetypes motivate the use of complementary CATFISH component tests because different methods are preferentially sensitive to different ranked signal shapes.

The distinction is not cosmetic. The ACAT operator maps each component p-value through $\tan\{(1/2-p)\pi\}$, so a value at the upper boundary contributes a large negative term; a component that could not be computed would otherwise vote against the pathway and bias the omnibus toward the null. The rule is applied per pathway rather than by set size, because a large pathway in which no gene is individually strong is equally uncomputable. It does condition the component on having at least one gene below a threshold, so where TFisher does contribute its p-values are stochastically small; because the resampling null is generated under the identical rule, that conditioning appears in the null distribution and is absorbed by the calibration. An analytic omnibus p-value under this rule would be anti-conservative, whereas the resampled one is not.

Precision limit. The soft TFisher implementation returns the null distribution function of the soft-thresholding statistic, so the one-sided p-value is formed as one minus that value. In double precision the difference is exactly zero once the distribution function exceeds $1-10^{-16}$, and a strongly enriched pathway would then report $p=0$. We cap the soft TFisher p-value at 10^{-16}, the smallest value the computation can resolve, rather than at the general lower floor applied to components whose tails are computed in closed form. A cap below the precision limit asserts resolution the statistic does not have; Section 4.3 shows what that does to the family-wise threshold.

2.2 Two sources of dependence

CATFISH statistics are correlated at two levels. The two arise from different causes, and both have to be addressed before any pathway p-value means what it claims to.

Within a pathway, component tests are dependent for two reasons: each is a deterministic function of the same $\mathbf{p}_S$, inducing cross-method dependence even if genes were independent; and gene-level evidence is correlated via LD, overlapping SNP-to-gene assignments, and shared local genomic architecture. Thus, any analytic reference distribution assuming independence is wrong for every component and typically anti-conservative, because correlated genes contain less information than the same number of independent genes.

Between pathways, pathway-level p-values are correlated because databases assign genes to multiple sets. A gene shared across pathways propagates its (possibly extreme) null draw to all of them, shifting their evidence in the same direction. The strength and pattern of this dependence depend on database structure: hierarchies with nested sets create widespread but weak dependence, whereas largely disjoint curated collections create rare but strong dependence among overlapping sets. Consequently, independence-based multiple-testing corrections (Bonferroni or Benjamini–Hochberg under its positive-dependence conditions) need not achieve nominal error rates; the discrepancy is driven by the database, not the method.

Table 2 CATFISH pathway-level component statistics. For a pathway S with $m = |S|$ genes and gene-level p-values p_g, each component emphasizes a different pathway signal pattern. The analytic reference distributions define the statistics; final inference uses the dependence-aware calibrations of Section 2.3.

Component	Pathway statistic	Analytic reference	Primary sensitivity
ACAT	$T_{\mathrm{ACAT}}(S) = \frac{1}{m}\sum_{g\in S}\tan\{\pi\,(\frac{1}{2} - p_g)\}$	$p_{\mathrm{ACAT}}(S) = \frac{1}{2} - \frac{1}{\pi}\arctan T_{\mathrm{ACAT}}(S)$	Sparse or extreme-tail signals.
Fisher	$T_{\mathrm{F}}(S) = -2\sum_{g\in S}\log p_g$	Under independence, $T_{\mathrm{F}} \sim \chi^2_{2m}$.	Broad enrichment with many small-to-moderate p-values.
Soft TFisher	$W(S;\tau) = \sum_{g\in S}[-2\log p_g + 2\log\tau]_+$	Adaptive over a τ grid; minimum calibrated p-value is taken.	Intermediate signals concentrated in a subset of genes.
Stouffer	$Z_{\mathrm{St}}(S) = \frac{1}{\sqrt{m}}\sum_{g\in S} Z_g, \quad Z_g = \Phi^{-1}(1-p_g)$	$p_{\mathrm{St}}(S) = 1 - \Phi(Z_{\mathrm{St}}(S))$.	Weak, diffuse, pathway-wide enrichment.
minP	$T_{\min}(S) = \min_{g\in S} p_g$	Under independence, $p_{\min} = 1-(1-T_{\min})^m$.	Single strongest gene or highly sparse signals.

These are often treated as separate problems, but in CATFISH both reduce to the same object: a vector of correlated gene-level statistics. A single genome-wide null draw, carrying the estimated LD, is scored through all pathways at once. Within a pathway, it provides null components and accounts for within-pathway dependence; across pathways, it provides a null for the family minimum and accounts for between-pathway dependence. No correlation structure is estimated or modeled; both are inherited. Section 2.3 develops this.

The construction relies on three assumptions: (i) *Approximate normality*: under the null, the pathway-level vector satisfies $\mathbf{Z}_S \sim \mathcal{N}(0, R_S)$. (ii) *Correlation-matrix adequacy*: correlation matrices from MAGMA-reported gene pairs provide an LD-informed working approximation to the unobserved gene-level covariance; unreported pairs are set to zero, and adequacy is evaluated via calibration and robustness analyses below. (iii) *No residual structural confounding*: gene-level inputs are adjusted so pathway signals are not primarily driven by gene size or SNP density.

2.3 Calibration and family-wise control from a shared correlated null

CATFISH obtains two different p-values for every pathway: a marginal one, which asks whether that pathway is enriched, and a family-wise one, which asks whether evidence this strong would appear anywhere among the K pathways tested by chance. Both are read from a single set of resampled null draws. The procedure has four steps.

Step 1: generate the null.

Gene–gene correlations are read from MAGMA's `.genes.raw` output as a list of gene pairs and their correlations. For each chromosome c containing m_c genes we form an $m_c \times m_c$ correlation matrix R_c by starting from the identity and filling in the reported pairs,

$$R_c = I_{m_c} + \sum_{(i,j)\in\mathcal{P}_c} r_{ij}\left(e_i e_j^\top + e_j e_i^\top\right),$$

where $\mathcal{P}_c$ is the set of reported pairs for chromosome c; pairs not in $\mathcal{P}_c$ remain zero. We compute correlations only for gene pairs close enough to share LD, so R_c is banded by construction: zeros for distant genes are approximations, not imputations, whereas zeros for nearby but unreported pairs introduce approximation error (see Section 5). Genes on different chromosomes are treated as independent, which is structurally correct. If R_c is not positive definite, we floor eigenvalues at 10^{-6}, reconstruct, and renormalise the diagonal; the implementation reports affected chromosomes.

Writing $R_c = U_c^\top U_c$ for the Cholesky factor, replicate b draws

$$\mathbf{Z}_c^{(b)} = U_c^\top \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_{m_c}),$$

independently per chromosome, and the concatenation $\mathbf{Z}^{(b)}$ is one genome-wide null realisation carrying the estimated LD structure. Gene-level null p-values follow by the Gaussian copula $p_g^{(b)} = 1 - \Phi(Z_g^{(b)})$, matching MAGMA's one-sided sign convention. This is repeated for $b = 1, \ldots, B$.

We draw genome-wide, not pathway-by-pathway, because Step 4 requires all K pathways to be scored on the *same* realisation so shared genes perturb them jointly. Drawing each pathway from its own correlation submatrix would be marginally correct but would remove the dependence structure Step 4 needs.

Step 2: score every pathway on every replicate.

For each replicate we recompute all five component statistics and the omnibus operator $\mathcal{O}(\cdot)$ for every pathway in the family $\mathcal{F} = \{S_1, \ldots, S_K\}$, giving a $B \times K$ matrix of null omnibus values

$$Q_k^{(b)} = \mathcal{O}\left(\mathbf{p}^{(b)}(S_k)\right), \qquad b = 1, \ldots, B, \quad k = 1, \ldots, K.$$

Because all components are recomputed from the same draw, Q preserves within-pathway gene dependence, cross-method dependence, and—since all K pathways share the draw—between-pathway dependence. The observed values $q_k^{\mathrm{obs}} = \mathcal{O}(\mathbf{p}^{\mathrm{obs}}(S_k))$ are computed the same way from the real data. Applying $\mathcal{O}$ to the observed components without resampling yields the uncalibrated analytic omnibus $\mathrm{Omnibus}_{\mathrm{An}}(S_k) = q_k^{\mathrm{obs}}$, which assumes gene independence and serves as the baseline.

Step 3: the marginal p-value.

Compare each pathway's observed statistic with its own null values across the B replicates. The LD-aware marginal p-value is the Monte Carlo tail probability

$$\tilde{p}_k \;=\; \frac{1 + \sum_{b=1}^{B} \mathbf{1}\Big(Q_k^{(b)} \le q_k^{\mathrm{obs}}\Big)}{B+1}.$$

This quantity is computed from the genome-wide draw rather than from the pathway-specific submatrix R_{S_k}; because R_{S_k} is simply the R_c submatrix for that pathway's genes, both use the same correlation information, so nothing is lost. We call $\tilde{p}_k$ the per-pathway calibrated p-value and use it to rank pathways. It corresponds to what is often called MVN calibration; we avoid that label because the multivariate normal model is the generator of the null draws, not a separate calibration step, and the same draws serve Step 4.

Step 4: the family-wise p-value.

Within each replicate, record the smallest statistic across the whole family,

$$M^{(b)} = \min_{1 \le k \le K} Q_k^{(b)},$$

and compare each observed statistic with the resulting distribution. This gives the Westfall–Young adjusted p-value (Westfall and Young, 1993)

$$\tilde{p}_k^{\mathrm{WY}} \;=\; \frac{1 + \sum_{b=1}^{B} \mathbf{1}\big(M^{(b)} \le q_k^{\mathrm{obs}}\big)}{B+1},$$

Rejecting all pathways with $\tilde{p}_k^{\mathrm{WY}} \le \alpha$ controls the family-wise error rate at level α under the complete null, whatever the dependence among pathways, because a false rejection anywhere in the family requires the family minimum to cross the threshold. Three further points apply. First, the control is in the strong sense, not only under the complete null. Westfall–Young procedures control the family-wise error rate strongly when the joint distribution of the statistics for the true null hypotheses does not depend on which other hypotheses are false—the condition of subset pivotality (Westfall and Young, 1993). Two features of the construction give it. Each pathway statistic is a deterministic function of its own genes' z-scores only; and under the model of Step 1, with association represented as a shift in the means of the associated genes, the genes of a pathway containing no associated gene are jointly $\mathcal{N}(0, R)$, restricted to those genes, whatever happens elsewhere. The observed statistics of the null pathways therefore have exactly their complete-null joint distribution, and because the minimum over any subset of pathways is never smaller than the minimum over the whole family, the probability that any null pathway crosses the threshold is at most α. The argument requires the soft TFisher truncation grid to be fixed in advance: a grid chosen from the empirical distribution of all gene-level p-values, as in data-adaptive implementations, would make every pathway's statistic depend on genes outside it. Second, the null is parametric: it is generated from the multivariate normal model of Step 1 rather than by permuting phenotypes, so the procedure is Westfall–Young in construction but inherits the adequacy of that model. Third, the counting form of the adjusted p-value above, in which null minima equal to the observed value count as at least as extreme, is required rather than a matter of taste. A threshold taken as a quantile of M and applied with an inclusive comparison rejects every observation tied at that quantile, and if the omnibus loses resolution in the tail so that many null minima coincide, the empirical family-wise error rate can approach one (Section 4.3).

Between-pathway correlation is never explicitly estimated or modelled. Because all K pathways are evaluated on the same draw in Step 2, genes shared across pathways perturb them jointly, as in the observed data, and this dependence is reflected in the distribution of M. The threshold $\hat{t}_\alpha = \mathrm{quantile}_\alpha(M)$ is the family-wise significance cutoff implied by the pathway database in use; its ratio to the Bonferroni cutoff α/K quantifies the database's departure from independence at no extra cost, provided it is read on calibrated statistics; on the analytic scale the ratio also absorbs any miscalibration of the per-pathway statistic (Section 4.1).

Steps 3 and 4 use the same matrix Q and require no additional resampling. Their outputs relate as a raw and adjusted p-value: $\tilde{p}_k$ assesses pathway S_k alone, while $\tilde{p}_k^{\mathrm{WY}}$ accounts for testing K pathways. Both are reported; neither replaces the other.

Size standardisation.

A minimum across tests is meaningful only when the tests share a scale. The raw omnibus does not: Fisher's statistic increases with m, so the Step 4 family minimum would mechanically favor the largest sets. We therefore also report a variant that first standardizes each observed value within its own column of Q as a z-score, $z_k = (q_k^{\mathrm{obs}} - \mu_k)/\sigma_k$, where μ_k and σ_k are that column's mean and standard deviation, and then takes the row minimum. This rescales values for the minimum operation; it is not a pathway-size bias correction, and Section 4.1 shows it does not behave as one. In the simulations of Section 2.6 every pathway in a configuration has the same size and dependence structure, so the pathways are exchangeable, share one marginal null, and the row minimum is taken on the raw scale; standardisation is needed only when pathway sizes differ, as in real data. We standardise by location and scale rather than by each pathway's empirical null distribution function. The latter would put pathways on an exactly common scale, but would resolve each to only $1/(B+1)$, which is coarser than the family minimum requires and would reintroduce ties.

2.4 Numerical implementation in log space

The omnibus statistic must be computed entirely in log space. This is a correctness requirement, not an optimisation, and it prevents silent failure.

The smallest positive normalized double is $\approx 2.2 \times 10^{-308}$. In a GWAS with $n \approx 1.3$ million, gene-level $p < 10^{-300}$ is common: in the GLGC LDL-cholesterol data (Section 3.9), the published p-value is exactly zero for 473 chromosome 1 variants, including the *SORT1* lead rs12740374 at $-\log_{10} p = 1725$. Combining even a few such genes with Fisher's method yields a pathway statistic whose double-precision p-value underflows to zero. All affected pathways then share the same floor value, ties are broken by storage order, and the ranking the analysis aims to produce is silently corrupted. In our

first implementation, the top-ranked pathways were (on inspection) alphabetical.

Five changes are needed. (i) Compute all component statistics on $\ell_g = \log p_g$ and return log p-values via `stats::pchisq(..., log.p=TRUE)` and `stats::qnorm(..., log.p=TRUE)`. (ii) Evaluate the ACAT sum using the small-p asymptote: for $p < 10^{-15}$, use $\tan\{\pi(\frac{1}{2} - p)\} \approx 1/(p\pi)$, accumulating those terms as a log-sum-exp of $-\ell_g - \log \pi$; use the trigonometric form for the rest. (iii) Replace $p = \frac{1}{2} - \arctan(T)/\pi$ with $p \approx 1/(T\pi)$ for $T > 10^6$, where cancellation near $\frac{1}{2}$ dominates truncation error. (iv) Since ACAT can be negative when most inputs exceed 0.5, handle that case in a separate branch computing $\frac{1}{2} - \arctan(T)/\pi$ directly; clamping would set $p = 0.5$ and destroy ordering in the upper half. (v) The minP inversion $1 - (1 - q)^m$ underflows for $q \lesssim 10^{-16}/m$, causing minP to drop out; use the mq asymptote whenever $\log m + \ell_{\min} < \log 10^{-8}$.

These changes let the pipeline represent pathway p-values down to $e^{-745} \approx 10^{-324}$ and, crucially, preserve their *ordering* well below where probability-scale computations fail. They are implemented in the scripts used for the real-data analyses. The simulations of Section 2.6 instead call the component functions of the CATFISH package directly, so that they validate the code a user runs; those functions work on the probability scale with a lower floor of 10^{-300}, which is adequate there because no simulated null approaches it. Soft TFisher is subject to its own, much higher, precision limit (Section 2.1).

2.5 Candidate-gene prioritization

The output a biologist acts on is a short list of genes, so CATFISH ranks genes by a composite score over three evidence layers:

$$\mathrm{Score}(g) = S_{\mathrm{GWAS}}(g) + S_{\mathrm{MAGMA}}(g) + 0.5\, S_{\mathrm{PATH}}(g) + 0.3\, B_{\mathrm{hub}}(g).$$

Each layer enters by rank rather than by its raw statistic, $S_\bullet(g) = -\log_{10}(r_\bullet(g)/N_\bullet)$, where r_{GWAS} is the gene's rank by minimum SNP-level p-value, r_{MAGMA} its rank by gene-level p-value, and r_{PATH} the rank of its best pathway; rank one is the strongest evidence. The hub term $B_{\mathrm{hub}}(g) = \log_{10}(n_{\mathrm{path}}(g) + 1)$ rewards membership in several pathways. Genes missing a layer score zero for it, so an absent layer neither helps nor penalises.

We use ranks instead of $-\log_{10} p$ to make layers with different tail behaviour comparable and to avoid the p-value floors in Section 2.4. Pathway evidence is halved because the pathway p-value is a set-level statistic shared across member genes; the hub term is intentionally secondary. We fixed weights *a priori*; simulations showed rankings were stable for $w_{\mathrm{path}} \in [0, 1]$ and $w_{\mathrm{hub}} \in [0, 0.5]$, with defaults on a broad plateau rather than a fragile optimum (**?**). We did not re-tune on the validation genes in Section 3.9.4 to avoid circularity.

The score is a heuristic ranking metric, not an inferential model: it has no null distribution or p-value. We also report per-layer ranks, because the GWAS and MAGMA layers use the same variants and are not independent evidence.

2.6 Simulation design

Simulations are run on the omnibus statistic under its null, except where a signal is injected deliberately (item 6). Two dependence designs are used. In the *shared-universe* design, pathways are overlapping subsets of one gene universe: $G = 3000$ genes in independent blocks of 50 with exchangeable within-block correlation ρ, standing in for chromosomes and LD blocks, from the first 600 of which we draw $K = 150$ pathways of m genes each at random. Each gene therefore belongs to several pathways, a pathway spans several blocks, and a single genome-wide draw moves every pathway containing a given gene—the between-pathway dependence the family-wise procedure is designed to control. In the *exchangeable* design, used only in item 1, each pathway is drawn independently and all m of its genes are mutually correlated at ρ. That design has no between-pathway dependence; it isolates within-pathway correlation at an intensity well beyond what block-structured LD produces, and serves as a stress test. Each study reports two summaries of the null matrix: per-pathway marginal type I error (by columns) and family-wise error rate (by rows), with Bonferroni as the family-wise comparator. Thresholds are fitted on one set of draws and evaluated on a held-out set, so rates are not correct by construction. Component statistics are computed with the CATFISH package functions, using the soft TFisher grid, undefined-component rule and precision cap of Section 2.1. We do not simulate genotypes or phenotypes or run a full GWAS; that is future work (Section 5) and is not needed for the questions here. Each study corresponds to a script in `catfish2/scripts/` and writes a CSV to `catfish2/results/` from which the tables and figures are generated.

1. **Null calibration and type I error** (`01_null_calibration.R`), exchangeable design. Under the global null we compare four decision rules: the analytic independence-based p-value; per-pathway calibration against a null bank shared by all pathways of a given size; Bonferroni; and the Westfall–Young adjusted p-value of Step 4, with the null distribution of the family minimum estimated on one half of the simulated families and evaluated on the other. We vary $\rho \in \{0, 0.3, 0.6\}$ and $m \in \{5, 20, 50\}$ and report type I error at $\alpha \in \{0.05, 0.01\}$, family-wise error, the genomic-control inflation factor $\lambda = \mathrm{median}(\chi^2_1\text{-transformed } p)/0.4549$, and the number of distinct values taken by the family minimum, which detects loss of resolution directly.
2. **Component-breakage stress test** (`02_breakage_stress.R`). We miscalibrate k of five components by replacing each affected component's p-value with p^γ, and test whether the omnibus and its calibration remain valid. We run both corruption directions: for $0 < p < 1$, $\gamma < 1$ yields $p^\gamma > p$ (conservative) and $\gamma > 1$ yields $p^\gamma < p$ (anti-conservative), the latter being most relevant for false positives. We use $\gamma \in \{0.7, 1.4\}$; sweep $k = 0, \ldots, 4$; evaluate up to three component combinations per k; and cross with the same LD regimes at $m \in \{20, 50\}$. We apply corruption to both null replicates and observed statistics, allowing resampling calibration to absorb it.
3. **Leave-one-out component sensitivity** (`03_adaptive_loo.R`). We remove each component in turn, recompute the null bank and the calibration from scratch, and measure the change in type I error and λ. This isolates how much each component contributes to, or destabilises, the calibrated omnibus, and identifies components whose removal would change inference.
4. **Per-component calibration** (`09_component_calibration.R`). Rather than combining, we calibrate each of the five components against its own null and report analytic and calibrated type I error, family-wise rate and λ per component. This identifies which statistics LD actually breaks and whether a single

calibrated component is usable in place of the omnibus when the signal architecture is known in advance.

5. **Robustness to an incomplete correlation file** (`08_missing_correlation.R`). Data are simulated under the true correlation, but calibration (null bank, marginal p-values, and Westfall–Young threshold) uses a matrix where a fraction $\{0, 0.25, 0.5, 0.75, 1\}$ of off-diagonal entries is set to zero, mimicking unreported gene pairs. Evaluation draws are shared across missingness levels; only calibration varies. For real data, `10_pair_coverage.R` places a correlation file on this scale by reporting the fraction of within-pathway gene pairs on the same chromosome and close enough to be correlated for which MAGMA reports a value.
6. **Archetype power** (`17_archetype_power_wy.R`), shared-universe design with $\rho = 0.3$. One pathway in each family carries signal shaped as one of the five archetypes of Section 1.1; the remaining $K - 1$ are null. Signal is a mean shift on the gene-level z scale applied to a fixed fraction of the pathway's genes, at positions permuted in every replicate: SSS shifts 15% of genes by 3.0; MDS 60% by 1.1; WDE 90% by 0.55; MSA 10% by 1.8 and a further 60% by 1.1; and SGP 5% by 4.0, with at least one gene in every layer. Shifts are multiplied by an effect scale over a grid running from undetectable to saturated, at $m \in \{5, 20, 50\}$. For each component and for the omnibus we record the probability that the signal pathway clears its own Westfall–Young threshold. Regret is the difference in that probability between the best single component and the omnibus in the same cell.

Unless stated otherwise, each configuration uses B null draws for the calibration bank and a separate set of replicate datasets for evaluation, with both values set at the top of each script so that a reduced smoke-test configuration can be run with `bash run_all.sh quick`.

2.7 Real-data analyses

Maize grain nitrogen. Gene-level statistics were computed with MAGMA from a mixed-linear-model (MLM) GWAS of grain nitrogen content, using a 10 kb upstream and downstream window for SNP-to-gene assignment. Pathways were taken from CornCyc (Hawkins et al., 2021; Caspi et al., 2008), giving 172 pathways after size filtering. Gene–gene correlations were recovered per chromosome from MAGMA's `.genes.raw` output; we verified explicitly that all ten chromosomes were represented, after an earlier run in which only chromosome 1 correlations had been parsed and 85% of the genome was silently treated as independent. Candidate-gene scoring used the three-layer score of Section 2.5 over all 34,873 annotated genes.

Human LDL cholesterol. We used the Global Lipids Genetics Consortium LDL-cholesterol summary statistics for the European-ancestry meta-analysis ($n \approx 1.3$ million) (Graham et al., 2021).

Variant filtering and the p-value floor. Variants were restricted to rsIDs present in the 1000 Genomes European reference panel used for LD. The released `pvalue` column underflows to exactly zero at the strongest loci, so all probabilities were reconstructed from `pvalue_neg_log10`. For the MAGMA gene analysis, whose `--pval` input requires a probability, these were floored at 10^{-300}; MAGMA is unaffected because the floor is far beyond any gene-level resolution it reports. The candidate-gene GWAS layer is affected, because it ranks genes by their minimum variant p-value and the floor collapses the strongest genes into a single tie. That layer therefore reads an unfloored $-\log_{10} p$ file written by `11_munge_neglog10.sh`; Section 3.9 reports what the floor costs when it is not removed.

Gene analysis. Variants were assigned to genes with a 10 kb upstream and downstream window using the NCBI37.3 gene locations. MAGMA gene analysis was run per chromosome with `--batch` and combined with `--merge`, which preserves the gene–gene correlations in the merged `.genes.raw` file; running the chromosomes as independent jobs and concatenating the outputs would discard them.

Pathways. Pathway inference used the Reactome subset of MSigDB C2:CP (1,390 sets after size filtering), with HumanCyc additionally converted into the CATFISH pathway format for comparison. We report Reactome because a hierarchical database is where between-pathway dependence is strongest, which is the regime the family-wise procedure is designed for.

Candidate genes. The GWAS layer uses MAGMA's own `.genes.annot` file rather than re-deriving a window, so that the GWAS and gene layers are assigned from identical variant sets. The pathway layer is ranked on the observed omnibus statistic rather than the Westfall–Young adjusted p-value. The two give the identical ordering, since the adjusted p-value is a monotone transform of the statistic, but the adjusted value is floored at $1/(B+1)$ and the entire top of the list is tied there at the B used here.

Positive controls. Thirteen genes with established roles in LDL-C biology—*LDLR*, *PCSK9*, *APOB*, *APOE*, *SORT1*, *HMGCR*, *NPC1L1*, *ABCG8*, *LPA*, *CETP*, *LDLRAP1*, *MYLIP* and *ANGPTL3*—were named before the prioritization was run and are used to assess it. They are a validation set, not a discovery; the score has no access to them.

3 Results

3.1 Independence-based calibration is anti-conservative; per-pathway and Westfall–Young calibration are not

Under the global null, the analytic omnibus p-value is valid only if the genes within a pathway are independent. Table 3 and Figure 2 show what happens when they are not. At $\rho = 0$ and $m = 5$ the analytic type I error was 0.0571 against a nominal 0.05, which is the behaviour one expects when the generating assumption holds. As either the within-pathway correlation or the pathway size increases, the analytic calibration departs from nominal, reaching 0.3332 at $\rho = 0.6$, $m = 50$, the largest value in the grid. The direction is the expected one: correlated genes carry less independent information than the analytic reference distribution assumes, so the statistic looks more extreme than it is.

Per-pathway calibration against the correlated null removed most of this. Empirical type I error was 0.0396 at $\rho = 0$, $m = 5$ and 0.0484 at $\rho = 0.6$, $m = 50$, remaining within $[0.0386, 0.0653]$ across the whole grid. At the stricter level $\alpha = 0.01$, the analytic calibration reached 0.2997 against the calibrated statistic's 0.0155, indicating that the discrepancy grows rather than shrinks in the tail, which is where pathway analysis operates.

The family-wise columns test the Westfall–Young construction, and here they are a genuine test rather than a check of arithmetic: the null distribution of the family minimum is estimated on half of the simulated families and applied to the other half. Empirical family-wise error ranged from 0.0120 to 0.0680 across the grid, within the Monte Carlo error of 250 held-out families, for which the standard

Table 3 Empirical type I error and genomic-control inflation for the CATFISH omnibus under the global null, by within-pathway LD level and pathway size. Analytic: independence-based omnibus. Calibrated: per-pathway calibration against the correlated null. Bonferroni and Westfall–Young (WY): family-wise error rates, the probability of at least one rejection among the pathways of one simulated family. Nominal level $\alpha = 0.05$.

LD	ρ	m	Analytic	Calibrated	Bonferroni	WY	λ_{an}	λ_{cal}
independent	0.0	5	0.0571	0.0396	0.0526	0.0320	0.909	0.947
independent	0.0	20	0.0626	0.0534	0.0620	0.0400	0.992	0.943
independent	0.0	50	0.0651	0.0481	0.0658	0.0120	1.046	0.961
moderate	0.3	5	0.1026	0.0653	0.6806	0.0360	0.701	1.055
moderate	0.3	20	0.2115	0.0569	1.0000	0.0320	0.477	1.031
moderate	0.3	50	0.2895	0.0469	1.0000	0.0640	0.261	0.970
strong	0.6	5	0.1354	0.0403	0.9806	0.0600	0.486	1.124
strong	0.6	20	0.2622	0.0386	1.0000	0.0440	0.068	1.031
strong	0.6	50	0.3332	0.0484	1.0000	0.0680	0.002	1.016

error at the nominal level is about 0.014. Bonferroni, applied to the analytic p-value, inherits that value's miscalibration: its family-wise error was 0.0526 in the independent configuration with $m = 5$, where the analytic p-value is valid, but 1.0000 at $\rho = 0.3$, $m = 20$ and 1.0000 at $\rho = 0.6$, $m = 50$. The pathways in this design are independent of one another, so none of that inflation comes from between-pathway dependence; all of it is the per-test statistic being wrong, compounded across the family.

Reaching these numbers required two corrections worth recording, because both failures were silent. In an earlier version the family-wise error at $\rho = 0.6$, $m = 50$ was 1.00. The cause was not the Westfall–Young logic but a loss of resolution in the omnibus. Soft TFisher's tail underflowed for about 3.5% of null pathways in that configuration, each such pathway was assigned the same floor value, and with 200 pathways per family nearly every family's minimum was that one shared number; a quantile of a distribution concentrated on a single value, applied with an inclusive comparison, rejects every family. Capping soft TFisher at the precision it actually has (Section 2.1) and computing the adjusted p-value in its counting form (Section 2.3) restored the family minimum to a continuous statistic: even in the worst configuration it takes 500 distinct values across the 500 simulated families.

3.2 Genomic-control inflation tracks the same failure

The genomic-control inflation factor λ summarises the same null in a form familiar from GWAS practice (Figure 3), and it points the wrong way. For the analytic omnibus, λ was 1.046 at $\rho = 0$, $m = 50$ and fell to 0.002 at $\rho = 0.6$, $m = 50$—far *below* one—in exactly the configuration where the analytic type I error was largest. The two summaries are not in conflict. λ is computed from the median of the null p-values and type I error from their lower tail, and under strong within-pathway correlation the analytic null distribution is not shifted but reshaped: excess mass in the lower tail together with a median above one half. An analyst reading λ alone would conclude the test was conservative. After per-pathway calibration λ lay in $[0.943, 1.124]$. A pathway analysis reporting λ near or below one is therefore not evidence that its tail is calibrated.

3.3 The omnibus degrades gracefully when its components are broken

A multi-test omnibus is only useful if a single misbehaving component cannot dominate it, and a family-wise procedure is only useful if that robustness survives at the family level too. We tested both by corrupting components on purpose, replacing the p-value of each of k selected components with p^{γ} and running the conservative ($\gamma = 0.7$) and the anti-conservative ($\gamma = 1.4$) direction over the overlapping pathway universe of Section 2.6, in which each gene belongs to 8.8 pathways on average (Table 4, Figure 4).

With no components corrupted ($k = 0$), the analytic omnibus had mean marginal type I error 0.1049 and the calibrated omnibus 0.0501; at the family level the Westfall–Young rate was 0.0560 and Bonferroni 0.1407. Under anti-conservative corruption of four of the five components ($k = 4$, $\gamma = 1.4$) the analytic omnibus degraded to 0.1827, while the calibrated marginal rate held at 0.0500 and the Westfall–Young family-wise rate at 0.0553 against Bonferroni's 0.3430. Under conservative corruption the analytic omnibus moved the other way, to 0.0601, losing power rather than control, while the calibrated marginal and family-wise rates held at 0.0500 and 0.0562. Across both directions and every configuration the family-wise rate stayed within $[0.0380, 0.0787]$ and the marginal rate within $[0.0425, 0.0612]$. Genomic-control inflation for the same runs (Figure 5) tracks the analytic statistic's drift in both directions while staying near one for the calibrated statistic.

The interpretation is specific and worth stating carefully. Resampling calibration is robust to component miscalibration because the *same* corruption is applied to the null replicates as to the observed data, so a component that is uniformly miscalibrated shifts the null and the observation together and largely cancels; the family-wise threshold, being a quantile of the same corrupted null, moves with them. This is a real and useful property, but it protects against systematic miscalibration, not against a component that is wrong in a way that correlates with the signal of interest. A component whose error depended on the data would not cancel, and we have not tested that case. The analytic calibration has no such protection and inherits the corruption directly, in whichever direction it was applied.

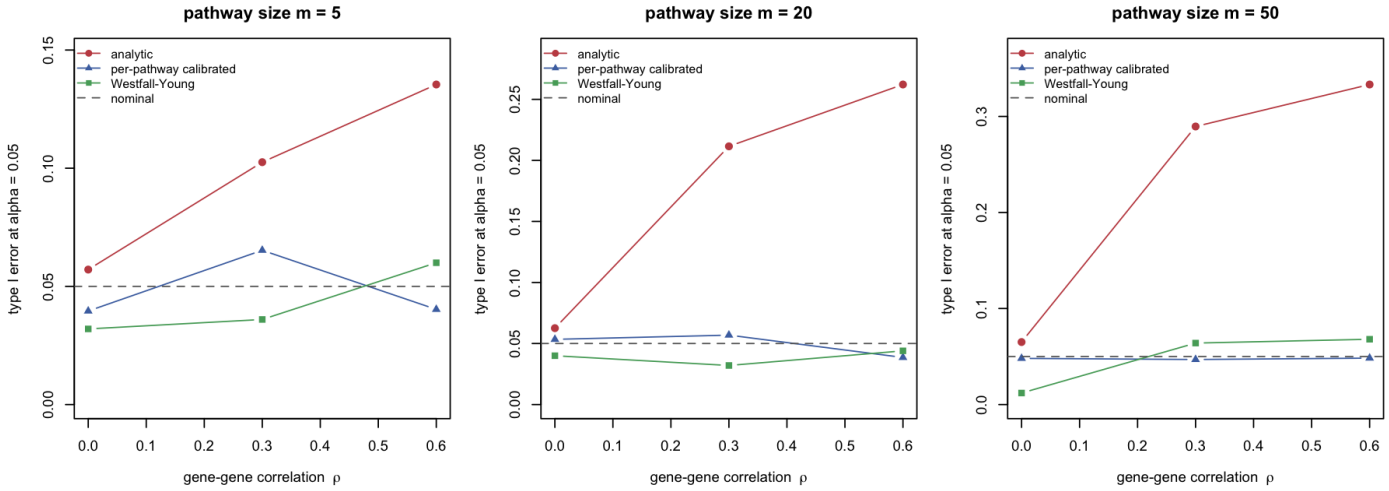


Figure 2 Empirical type I error of the CATFISH omnibus under the global null at $\alpha = 0.05$, by within-pathway LD level ($\rho = 0,\ 0.3,\ 0.6$) and pathway size ($m = 5,\ 20,\ 50$). The horizontal line marks the nominal level. The analytic independence-based calibration departs from nominal as either correlation or pathway size increases; per-pathway calibration against the correlated null, and the Westfall–Young family-wise error rate, both track the nominal level.

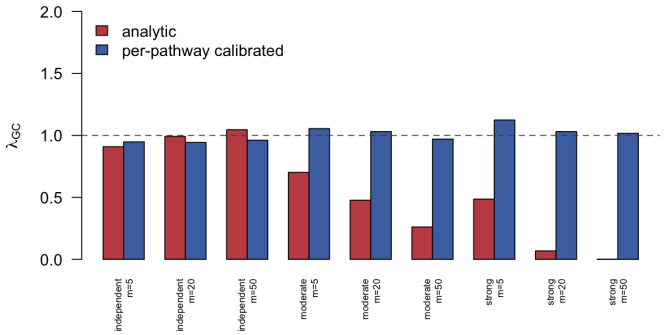


Figure 3 Genomic-control inflation factor λ for the analytic and per-pathway calibrated omnibus under the global null, by LD level and pathway size. The horizontal line marks $\lambda = 1$.

3.4 No single component is load-bearing

Removing each component in turn and rebuilding the null bank, the marginal calibration and the family-wise threshold from scratch (Table 5, Figure 6) left the marginal type I error within $[0.0400, 0.0625]$ against 0.0501 for the full five-component omnibus, and the Westfall–Young family-wise rate within $[0.0300, 0.0800]$ against 0.0560. The largest departure from nominal followed removal of `Stouffer` (0.0573).

The same table carries the sharpest contrast in the paper between the two family-wise procedures. Averaged over pathway sizes and leave-one-out variants, the Westfall–Young rate was 0.0670, 0.0428 and 0.0588 in the independent, moderate and strong LD regimes—flat at nominal—while the Bonferroni rate moved from 0.0545 to 0.1475 to 0.2137, reaching 0.2720 at worst. We are careful about what this does and does not show. Bonferroni is applied here to the analytic omnibus p-value, which is itself anti-conservative under LD (Section 3.1), so the inflation is the compounding of an uncalibrated per-test statistic with a correction that cannot repair it, which is what a pipeline that skips calibration actually does. In the independent rows, where the analytic p-value is valid, Bonferroni was close to nominal: overlap at an average of 8.8 pathways per gene is not enough to make it noticeably conservative. The lesson is that a family-wise correction inherits whatever miscalibration the per-test statistic carries, whereas the Westfall–Young threshold, being a quantile of the same null the statistic came from, does not.

Within that pattern, removing Stouffer, Fisher or soft TFisher lowered the Bonferroni rate, while removing ACAT or minP raised it. Stouffer and Fisher are the aggregation statistics that within-pathway correlation inflates (Section 3.5), and soft TFisher's uncalibrated value is a minimum over thresholds and therefore stochastically small; ACAT and minP stay close to nominal, so removing them leaves the omnibus more dominated by the inflated components.

No single deletion broke error control at either level, which is the expected consequence of calibrating empirically: the null bank is recomputed for whichever components remain, so both the marginal p-value and the family-wise threshold remain valid for any subset. What changes with the subset is sensitivity, not validity. The practical implication is that users may drop a component they distrust for a particular dataset—for example minP when gene-level p-values have been floored at a reporting limit—without invalidating the inference, at some cost in power against the architectures that component is best at.

Genomic-control inflation behaves the same way: the fourth panel of Figure 6 shows λ for the calibrated marginal statistic staying near one for every deletion, so no component is load-bearing for the inflation diagnostic either.

3.5 Linkage disequilibrium breaks the aggregation tests, not the extreme-value tests

The omnibus hides which of its inputs is failing. Calibrating each component separately (Table 6, Figure 7) shows that the damage LD does is concentrated in two of the five, and that the identity of those two is predictable from what the statistics measure.

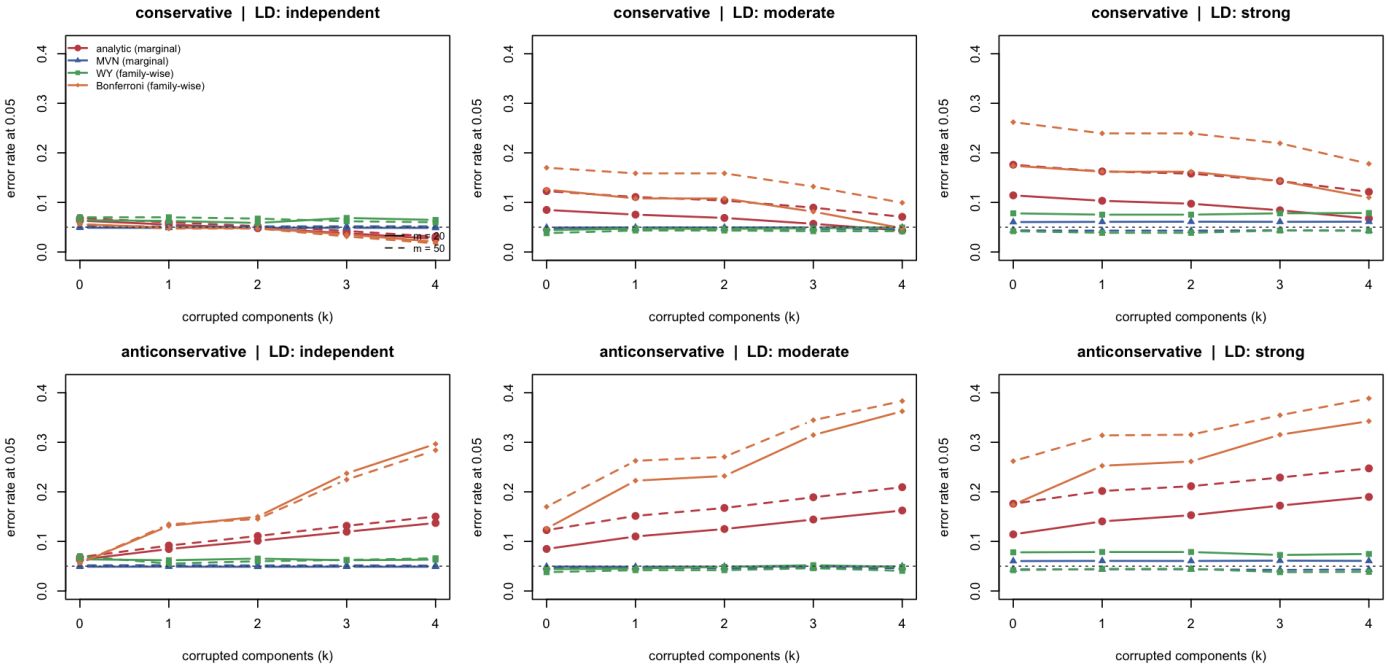


Figure 4 Component-breakage stress test on overlapping pathways. Rows are the direction of the deliberate corruption ($p \mapsto p^{\gamma}$, with $\gamma = 1.4$ anti-conservative and $\gamma = 0.7$ conservative); columns are the within-pathway LD level. Each panel shows the per-pathway marginal rates and the family-wise rates together, since all four are rates at the nominal 0.05 marked by the dotted line. Line type distinguishes pathway size. The analytic statistic and the Bonferroni correction applied to it both drift with k; the calibrated marginal rate and the Westfall–Young family-wise rate do not.

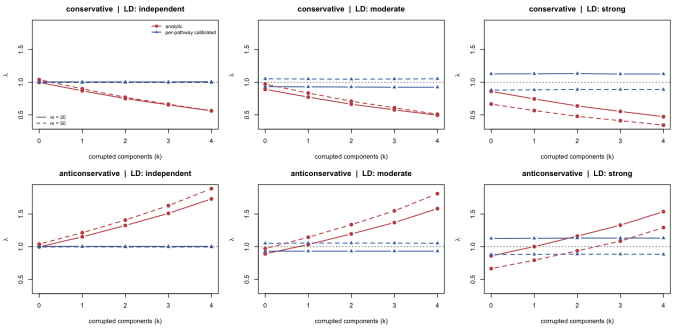


Figure 5 Genomic-control inflation λ for the component-breakage runs, by corruption direction and LD level. λ is a property of the marginal p-value distribution and has no family-wise counterpart, so it is shown separately from Figure 4. The analytic statistic drifts with k in whichever direction the components were corrupted; the calibrated statistic stays near one.

Under the independent null every component was correctly sized (for example ACAT at 0.0555 and Stouffer at 0.0511), which is the expected behaviour when the generating assumption holds. Under strong LD the two aggregation statistics—Fisher, which sums $-2\log p$ across the pathway, and Stouffer, which sums Z-scores—inflated to 0.1632 and 0.1887 against the nominal 0.05. The two extreme-value statistics barely moved: ACAT went to 0.0555 and minP to 0.0431. The mechanism is transparent. An aggregation statistic adds one term per gene and implicitly treats each as independent evidence, so correlated genes are counted repeatedly and the statistic grows; a statistic that depends on the most extreme gene is insensitive to how many near-duplicates of it the pathway contains. The omnibus inherited the inflation of its worst inputs, reaching 0.1761.

Calibration against the correlated null removed this for every component, bringing Fisher to 0.0411, Stouffer to 0.0501 and the omnibus to 0.0435. The practical consequence is that a user who knows a pathway's signal architecture may use a single calibrated component in place of the omnibus without losing error control, which is what makes the architecture-specific testing described in Section 2.3 usable.

This result also explains an observation in Section 3.4: removing an aggregation component lowered the Bonferroni family-wise rate while removing ACAT or minP raised it. The aggregation components are the ones that were inflated in the first place.

Three implementation notes belong here, all about soft TFisher. First, the statistic is undefined on a pathway in which no gene falls below any truncation threshold. Under the grid of Section 2.1 this did not occur at $m \geq 20$ in any configuration, but it is frequent in small pathways under grids confined to small τ, and a single undefined draw in a calibration bank makes that pathway's calibrated p-value undefined unless non-finite draws are filtered before the tail count, which the implementation does. Second, an undefined component must be excluded from the omnibus rather than entered at $p = 1$ (Section 2.1). Because ACAT weights by $\tan\{(1/2 - p)\pi\}$, entering the boundary value makes a test that could not be computed act as evidence against the pathway; under the null it pushes the median omnibus p-value above one half, so that λ falls below one while the tail is simultaneously inflated. Third, soft TFisher's uncalibrated value is a minimum over thresholds rather than a p-value, so its analytic columns in Table 6 are not interpretable as calibration; only its calibrated and family-wise columns are.

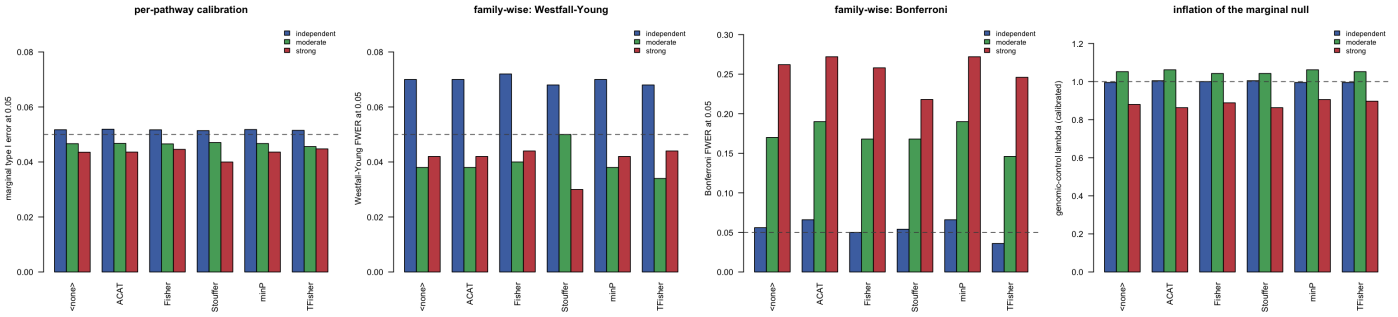


Figure 6 Leave-one-out component sensitivity at $m = 50$, with each component removed in turn and the null bank, marginal calibration and Westfall–Young threshold rebuilt from scratch. From left: marginal type I error, Westfall–Young family-wise error rate, Bonferroni family-wise error rate, and genomic-control inflation λ for the calibrated marginal statistic. The dashed line marks the nominal level (0.05, or $\lambda = 1$). Note the y-scale of the third panel, which is the point of the comparison.

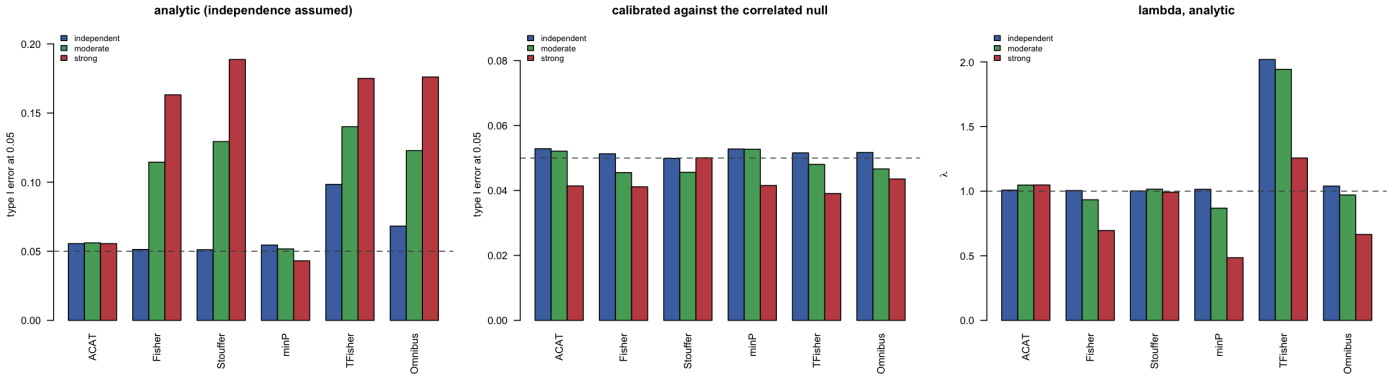


Figure 7 Per-component calibration at $m = 50$. Left: type I error at $\alpha = 0.05$ under the analytic independence assumption. Centre: the same components calibrated against the correlated null. Right: genomic-control inflation λ for the analytic statistics. The aggregation components, Fisher and Stouffer, are the ones LD inflates; the extreme-value components, ACAT and minP, are nearly unaffected. Calibration restores all five.

3.6 Both calibrations degrade when the correlation file is incomplete, and the family-wise one degrades faster

Every result above assumes the gene–gene correlation matrix is the one the data were generated from. In practice it is assembled by filling MAGMA's reported gene pairs into an identity matrix, so a pair MAGMA does not report is silently treated as independent. We measured the cost directly by generating data from the true correlation structure while calibrating from a matrix with a known fraction of its off-diagonal entries zeroed (Table 7, Figure 8).

With a complete matrix the marginal type I error was 0.0535 and the Westfall–Young family-wise rate 0.0525, both at nominal. Dropping half the pairs raised them to 0.0823 and 0.1225; dropping all of them, which is the independence assumption in its pure form, gave 0.1090 and 0.2075 under moderate LD and 0.1424 and 0.2200 under strong LD.

Expressed as multiples of each procedure's own nominal level—the only fair comparison between a per-pathway rate and a per-family one—the marginal calibration was 1.6× nominal at 50% missingness and the family-wise procedure 2.4×. Westfall–Young is therefore more sensitive to an incomplete correlation file than the marginal calibration it is built from. The reason is structural rather than incidental: the family minimum over K pathways is a tail statistic, so it depends on the extreme of the null distribution, which is precisely where an error in the correlation structure does the most damage. The marginal p-value depends only on the bulk.

This does not overturn the case for family-wise control, because the comparison that matters is against the correction it replaces rather than against the marginal rate it is not competing with. Bonferroni's family-wise rate in the same runs was 0.1975 and 0.2150 and does not depend on the correlation file at all, since it is computed from the analytic p-value. Westfall–Young remained the better family-wise procedure up to 75% missingness, and lost only when essentially no correlation information remained—at which point it is calibrating against a null that does not resemble the data.

It does, however, change how the procedure should be reported. A Westfall–Young p-value is only as trustworthy as the correlation file behind it, and the completeness of that file is measurable rather than assumed. The quantity to check is narrow: a correlation matrix from MAGMA is *supposed* to be sparse, because correlations are computed only for gene pairs close enough to share LD, so a zero between distant genes is correct rather than missing. What matters is the fraction of within-pathway gene pairs that are on the same chromosome and close enough to be correlated for which a value is actually reported. We provide `10_pair_coverage.R` for this, and recommend that it be run and its result stated alongside any family-wise p-value.

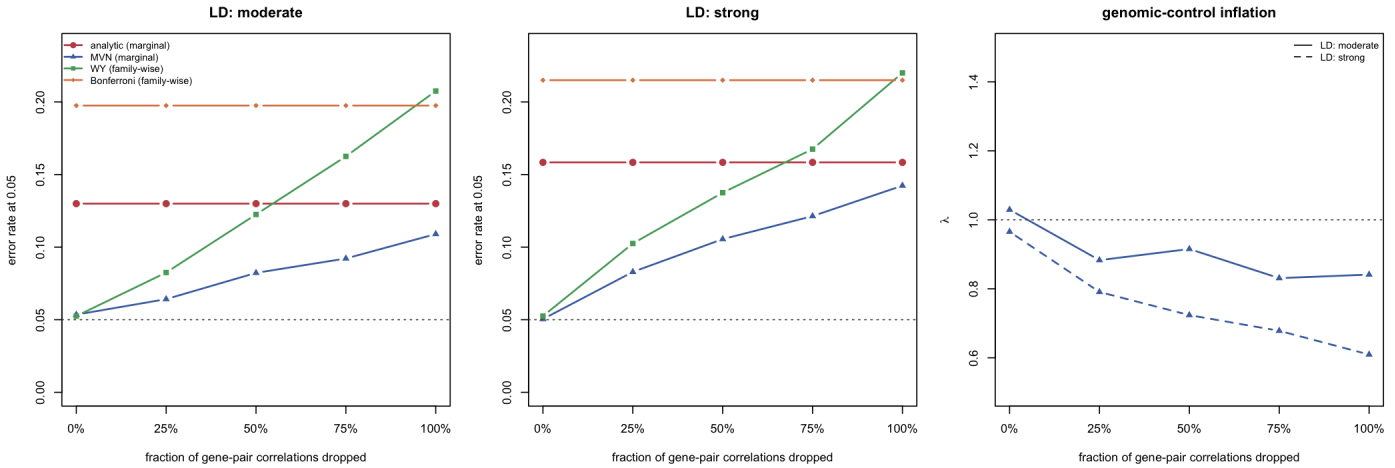


Figure 8 Robustness to an incomplete gene–gene correlation file. Data are generated from the true correlation structure while calibration uses a matrix with the stated fraction of off-diagonal entries zeroed. Left and centre: error rates at $\alpha = 0.05$ under moderate and strong LD. Right: genomic-control inflation λ for the marginal calibration, which falls below one rather than rising as the type I error more than doubles.

Finally, and consistently with Section 3.2, λ did not flag any of this. Across the missingness sweep it ranged from 0.609 to 1.029, drifting *below* one as correlation information was removed, while the marginal type I error more than doubled. This is the second independent setting in which λ at or below one coexists with an inflated tail, and it is the clearest argument we can make that λ should not be used as the sole calibration diagnostic in pathway analysis.

3.7 Each archetype is detected by the component predicted for it

Section 1.1 proposed five pathway signal archetypes and predicted, on statistical grounds, which component tests should be sensitive to each. That prediction is testable only under simulation, because the architecture of a real pathway is unknown. We generated pathways carrying each archetype's ranked signal shape, embedded them in a family of null pathways drawn from the same correlated gene universe, and measured the probability that the signal pathway clears the family-wise Westfall–Young threshold. Effect sizes were scaled over a grid so that every archetype passes from undetectable to saturated, and the component set, the universe and the resampling are exactly those used elsewhere in this paper.

The predictions hold. In 15 of 15 archetype-by-size cells the component that most often achieved the highest family-wise power was inside the set predicted in Table 1 (Table 8). The separation is sharp and in the expected direction: diffuse enrichment is recovered by Stouffer, with the extreme-tail tests near their null; single-gene and strongly sparse signals are recovered by minP and ACAT, with Fisher and Stouffer close to powerless; and dense or mixed architectures by Fisher, with soft TFisher close behind. No component is best everywhere, which is the premise the omnibus rests on.

The ranks below first place matter as much as the winners. Across the 36 informative cells, Fisher finished first 17 times but fourth or lower in 11, and minP finished last in 19. Soft TFisher finished first only 1 time, but second in 11 and third in 24, and never lower than third: it is the only component never among the weakest. Inside an omnibus that averages its inputs, that is the more useful property. A component that is never best but never near its null protects the combination precisely in the cells where the specialists disagree. It also qualifies Table 1: soft TFisher belongs among the tests sensitive to dense and mixed signal, where it ran close behind Fisher, but it was not the most sensitive test for any archetype.

What matters in practice is not which component wins but how much the omnibus gives up by not knowing in advance which one would. We measure that as *regret*: the difference between the best single component in a cell and the omnibus in the same cell. Regret is small for most architectures (Table 9, Figure 10), from 0.006 for MSA to SGP at 0.066. Against the alternative of choosing one component in advance, which risks the full gap between the best and worst component in a cell—frequently more than 0.5 in power—that is a favourable trade.

The exception is worth stating plainly. In the single-gene case the omnibus can give up as much as 0.476 in a single cell, because minP and ACAT detect the pathway while Fisher and Stouffer are near their null, and averaging across components dilutes a signal that only two of them can see. This is a property of combination rather than a defect of any component, and it is the price of not having to choose. It also marks the regime in which a gene-level test, rather than a pathway-level one, is the appropriate instrument.

3.8 Maize grain nitrogen: Westfall–Young recovers the known pathway and gene

We applied the full pipeline to a mixed-linear-model GWAS of maize grain nitrogen content, testing 172 CornCyc pathways. The leading pathway under the size-standardized Westfall–Young ordering was `HOMOSERSYN-PWY`, L-homoserine biosynthesis (13 genes; omnibus $p = 1.58 \times 10^{-3}$; size-standardized family-wise $p = 0.070$), followed by `PWY-5097`, L-lysine biosynthesis VI (25 genes; omnibus $p = 3.04 \times 10^{-3}$; size-standardized family-wise $p = 0.338$).

This ordering reproduces, independently and from a different statistical framework, the result of a previously reported analysis of the same trait (Tandukar and Rellán-Álvarez, 2025), in which L-homoserine biosynthesis was the only pathway enriched across all populations examined and L-lysine biosynthesis VI was the runner-up for reasons that analysis sets out. Recovering the same two pathways

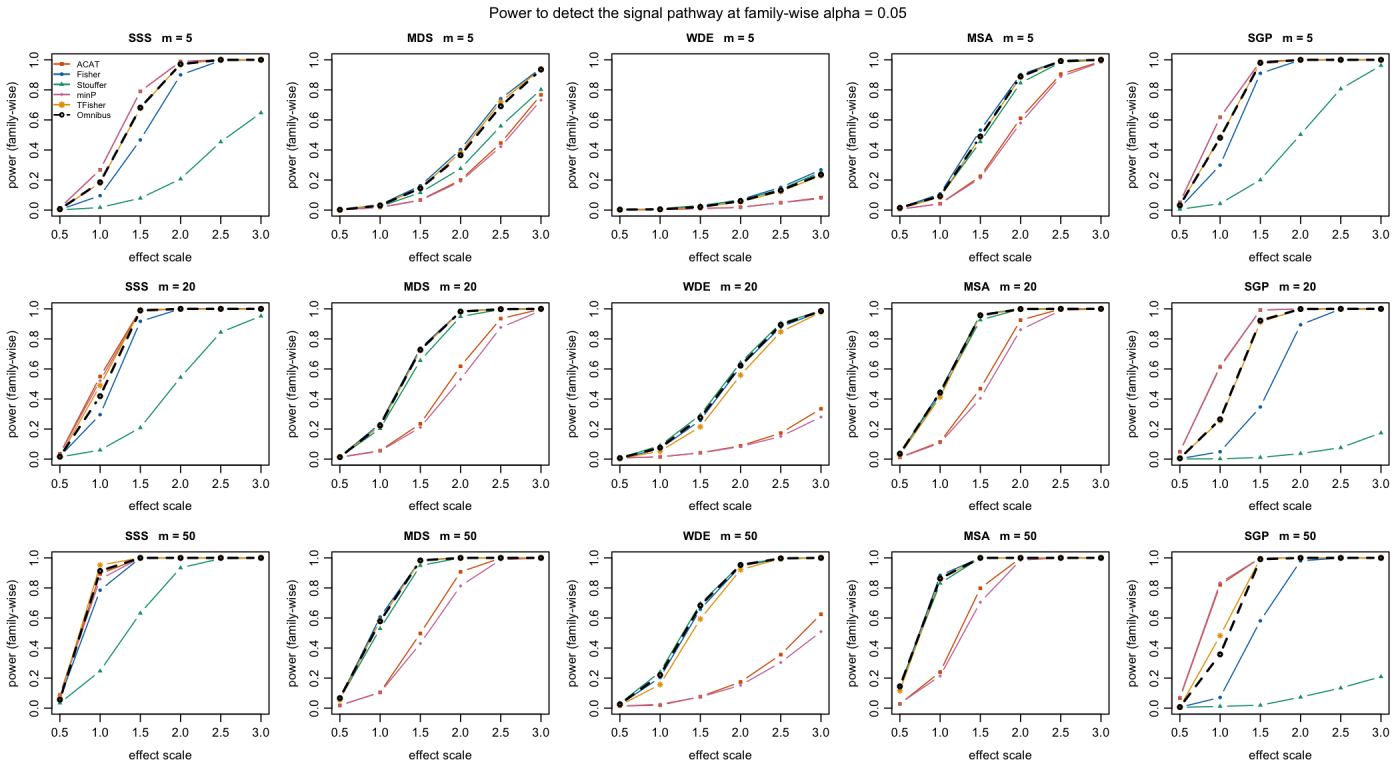


Figure 9 Family-wise power to detect a signal pathway, by archetype (columns) and pathway size (rows), as a function of effect scale. Each line is one component test; the dashed black line is the ACAT-O omnibus. The component predicted for each archetype in Table 1 separates from the others over the range in which power is informative.

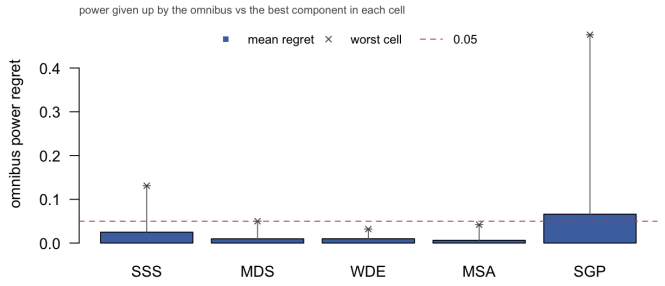


Figure 10 Power the omnibus gives up relative to the best single component in each cell. Bars are the mean across sizes and effect scales; crosses mark the worst single cell. The dashed line marks a regret of 0.05.

in the same order out of 172 candidates, using a method that shares no calibration machinery with the original, is the strongest external check we can presently make on the pipeline.

The three-layer candidate score of Section 2.5, computed over all 34,873 annotated maize genes with pathway evidence taken from the Westfall–Young adjusted p-values, placed `Zm00001eb220040` first. This gene is the aspartate kinase isozyme named in the earlier analysis, and L-homoserine biosynthesis was its highest-scoring pathway, so the gene- and pathway-level layers agree rather than merely coexisting.

Two caveats are essential. First, nothing in this analysis is significant: the leading pathway's size-standardized family-wise p-value is 0.070, and under the unstandardized Westfall–Young ordering it is 0.284 against a Bonferroni value of 0.272. The result is a ranking that agrees with prior biology, not a discovery. Second, grain nitrogen is highly polygenic and the MLM GWAS contained no genome-wide significant genes at all (Table 11), which is precisely the regime in which a ranking is the only available output. We regard the agreement as encouraging evidence that the ordering is meaningful in the low-signal regime, and we do not claim more than that.

3.9 Human blood lipids: saturation at the pathway level, intact resolution at the gene level

The GLGC lipid meta-analyses ($n \approx 1.3$ million) are the opposite regime from the maize analysis: very large samples, well-characterised traits, and gene-level signal everywhere. We analysed LDL-C and HDL-C through an identical pipeline—same reference panel, same gene annotation, same Reactome sets, same window, same number of permutations—so that the only thing differing between them is the phenotype. Together they produce one clearly negative result and one clearly positive one, and the pair says more than either alone.

3.9.1 The self-contained null is false almost everywhere

At the gene level, 44.1% of tested genes had $p < 0.05$ for LDL-C and 11.2% reached genome-wide significance; for HDL-C the figures were 52.7% and 14.8%, with a median gene-level p-value of 0.038. The consequence for self-contained pathway testing is arithmetical

rather than statistical. A *randomly chosen* set of fifty genes contains on average 5.6 genome-wide significant genes under LDL-C and 7.4 under HDL-C, and contains at least one with probability above 0.99 in both. The self-contained null hypothesis—that the genes in a set carry no more association than the background—is therefore false for essentially any set of genes one could assemble, including sets with no biological coherence whatever.

The output reflects this. Of the 1,390 Reactome pathways, 1,366 were significant under Westfall–Young family-wise control at $\alpha = 0.05$ for LDL-C and 1,384 for HDL-C. The natural first explanation is an inadequate multiple-testing correction, and we can rule it out: the Westfall–Young procedure controls the family-wise error rate exactly under the realized between-pathway dependence, and it removed essentially nothing. The rejections are not false positives in the sense a correction addresses. The null really is false for those sets. The problem is the hypothesis, not the threshold.

A second, subtler consequence is that the ranking acquires a strong dependence on pathway size. The twenty highest-ranked pathways had a median of 159 genes for LDL-C and 182 for HDL-C, against a median of 29 across all 1,390 sets, and the correlation between $\log_{10}$ set size and the pathway statistic was -0.825. In a saturated genome a large set samples more of an enriched background, so it accumulates more evidence for a reason that has nothing to do with the trait. Any interpretation of the ordering has to be made against that gradient, and we return to it below.

3.9.2 Two traits, different genes, the same pathway

LDL-C and HDL-C are biologically distinct arms of lipoprotein metabolism, and their gene-level results differ accordingly (Figure 11A). Across the 18,563 genes analysed in both, the Spearman correlation of gene-level Z statistics was 0.393. Of the top hundred genes in each trait, 13 were shared, and of the genes reaching genome-wide significance in either, the Jaccard overlap was 0.208 (830 shared of 3,987). These are not two readings of the same signal.

Despite that, the highest-ranked pathway of the 1,390 tested was the same in both: `REACTOME_PLASMA_LIPOPROTEIN_ASSEMBLY_REMODELING_AND_CLEARANCE`. At 74 genes it is well below the 159–182 median size of the surrounding top twenty, so it reaches first place against the size gradient of Section 3.9.1 rather than because of it.

The reason the two traits converge is visible in which genes carry the pathway in each (Figure 11B). Among its 74 members, the strongest LDL-C contributions come from the receptor-uptake arm—*LDLR* ($p = 3.1 \times 10^{-24}$), *PCSK9* (7.8×10^{-23}), *LDLRAP1* (7.4×10^{-22}), with *APOB*, *APOE* and *MTTP* close behind—while for HDL-C they come from the reverse-transport and lipolysis arm: *ABCA1* (1.3×10^{-33}), *LIPC* (5.8×10^{-29}), *SCARB1* (1.9×10^{-22}), *LPL* (8.9×10^{-22}), with *APOA1*, *ANGPTL3* and *LMF1* following. Only five of the fifteen leading drivers are common to both traits.

This is the behaviour a pathway method exists to produce. Two GWAS that disagree about which genes matter are nonetheless aggregated onto the same biological process, because that process contains both arms. The convergence is not a restatement of a shared gene-level signal; the gene-level signals are largely disjoint.

Two qualifications belong with the claim. First, the two traits are not independent replicates at the pathway level: across all 1,390 sets the Spearman correlation of the pathway statistic was 0.863, and 75 of the top hundred pathways were shared (Figure 11C). That correlation is driven by the size-dependent background of Section 3.9.1, which is common to both traits, rather than by the leading hits—which is precisely why the driver-level comparison, not the aggregate correlation, is the evidence for the claim above. Second, the phenotypes themselves are genetically correlated to a degree we have not estimated here; the gene-level Spearman of 0.393 is the quantity that actually enters CATFISH and is reported in its place.

Below the first rank, large generic sets—sumoylation, Golgi traffic, cellular response to chemical stress, rRNA processing—are interleaved with the lipid pathways in both traits, again consistent with the size gradient. Lipid, lipoprotein and sterol sets nonetheless concentrate near the top of both orderings, and they move between traits in a biologically sensible way: PPARα-regulated lipid metabolism ranks 16th for LDL-C and 7th for HDL-C, while sphingolipid metabolism ranks 81st and 182nd respectively.

3.9.3 The adaptive TFisher component is defensible but not load-bearing

The analyses above use all five component tests. Because the soft TFisher component is adaptive over a grid of truncation thresholds and is by some margin the slowest to evaluate, we also ran the complete HDL-C analysis with it removed, at matched $B = 10{,}000$, as a direct ablation.

The two orderings are almost identical: the Spearman rank correlation between the four- and five-component results was 0.9994, the median rank change was three places, and the largest change over all 1,390 pathways was 104 places, with 44 sets moving more than fifty. Every lipid pathway moved slightly upward when TFisher was included—plasma lipoprotein assembly $2 \to 1$, PPARα $8 \to 7$, lipoprotein clearance $98 \to 87$. This matches the leave-one-out simulation of Section 3.4, where removing TFisher left the family-wise error rate unchanged at 0.042.

We report this because the adaptive grid is the component most open to the objection that taking a minimum across five thresholds without correcting for the search is anti-conservative. That objection is correct for the analytic component—its uncalibrated type I error reaches 0.1750 at strong LD—but the resampling applies the same grid search to every null draw, so the inflation cancels, and the calibrated component is the best behaved of the five (0.0391). The ablation confirms that nothing in the real-data conclusions depends on it either way.

3.9.4 Gene prioritization is unaffected

The three-layer score of Section 2.5 was computed over all 18,563 genes. Table 10 gives the top five and Supplementary Table 12 the top fifty.

LDLR ranked 1 and *PCSK9* 3—the receptor that clears LDL from plasma and the protease that degrades it, the two genes around which LDL-lowering pharmacology is built. *APOE* and *APOC1*, at the chromosome 19 apolipoprotein cluster, follow close behind. Three of the top five are therefore genes with direct causal roles in LDL-C biology.

The other two require care, because they illustrate what a gene-based test can and cannot do. *PVRL2* (chr19:45.35–45.39 Mb) lies 17 kb from *APOE*, and *SMARCA4* (chr19:11.07–11.17 Mb) lies 27 kb from *LDLR*; *KIAA1324* (chr1:109.66–109.75 Mb), just outside the top five, sits in the *SORT1*/*CELSR2*/*PSRC1* region. None of these is a lipid gene. Each is an LD neighbour of one, and ranks highly because the association signal at a locus is shared across the genes within it. This is the expected behaviour of every gene-based test,

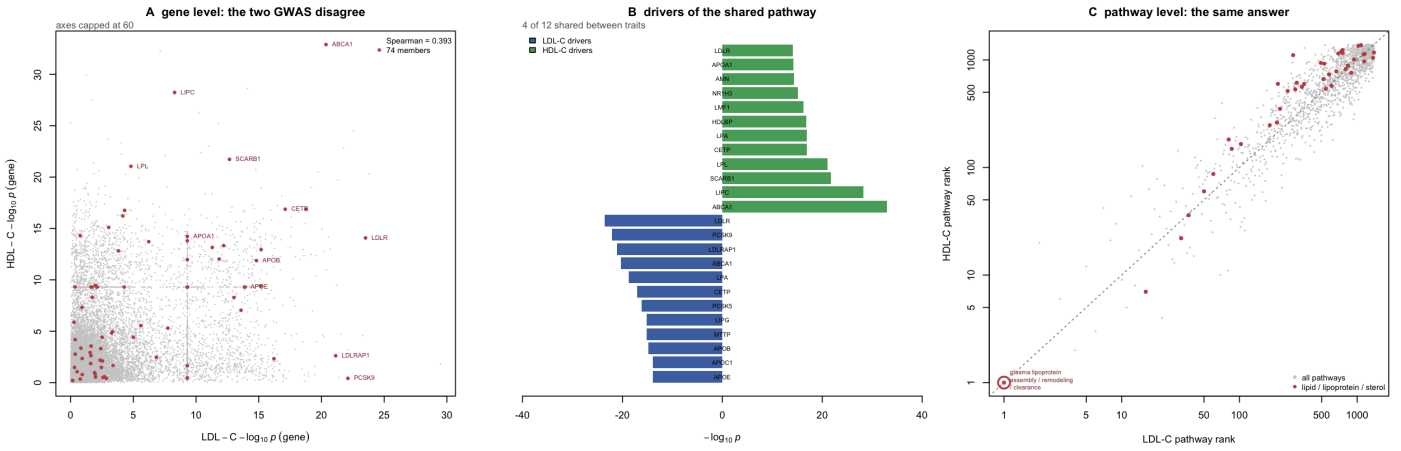


Figure 11 LDL-C and HDL-C reach the same pathway from different genes. **A** Gene-level $-\log_{10} p$ for the 18,563 genes analysed in both traits; the 74 members of the top-ranked pathway are highlighted and selected drivers labelled. The two GWAS agree only moderately (Spearman 0.393; 13 of the top hundred genes shared). **B** Within that pathway, the twelve strongest contributors in each trait. LDL-C is carried by the receptor-uptake arm and HDL-C by the reverse-transport and lipolysis arm, with five of fifteen leading drivers in common. **C** Pathway ranks in the two traits, lipid-related sets highlighted and the shared top pathway circled; it ranks first in both. The broad agreement away from the top reflects the size-dependent background of Section 3.9.1 rather than shared leading signal.

CATFISH included: the method ranks *loci*, and resolving the causal gene inside a locus requires evidence it does not use. We state this explicitly because a table of top-ranked genes invites the opposite reading.

Figure 12 shows where each of the top twenty genes sits in each layer, which separates genes carried by a single layer from those with concordant support across all three.

As a pre-specified check, all thirteen established LDL-C genes named in Section 2.7 were recovered, with a median rank of 20 out of 18,563 against a random expectation of 9,282, and 12 of 13 in the top 1%.

The informative detail is *which* layer produces this. The pathway ranks of the control genes are good but unremarkable—*LDLR* ranks 462nd of 9,727 pathway-annotated genes—while their GWAS and MAGMA ranks are near the top. The ordering is driven by the gene-level layers, and the pathway layer contributes little. That is the saturation result of Section 3.9.1 seen from the other side: saturation degrades the pathway layer specifically, because the self-contained hypothesis has stopped discriminating between gene sets, while leaving gene-level resolution intact. A composite score that weights pathway evidence at half the strength of direct genetic evidence degrades gracefully in exactly this regime, which was the reason for the weighting (Section 2.5.3) but not a situation it was designed against.

One implementation point affects this table directly and generalises beyond it. When the GWAS layer is ranked on probabilities reconstructed from GLGC and floored at 10^{-300}, 0 genes tie at the floor—including *LDLR*, *PCSK9*, *APOB*, *APOE*, *LPA* and *HMGCR*—and the layer cannot order the very genes it most needs to. Ranking on unfloored $-\log_{10} p$ removes the tie at no other cost. This is the same double-precision limit as Section 2.4, surfacing in a different part of the pipeline, and it is worth checking wherever a ranking is built from large-n GWAS summary statistics.

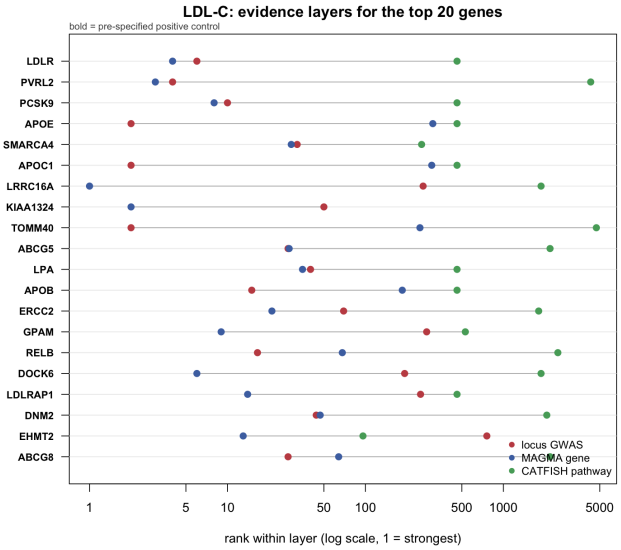


Figure 12 Evidence layers for the twenty highest-scoring genes in the LDL-C analysis. Each row is a gene; the three points give its rank within the locus-level GWAS layer, the MAGMA gene-level layer and the CATFISH pathway layer, on a log rank scale with one the strongest. Genes in bold are the pre-specified positive controls. The genetic layers place the controls near the top while their pathway ranks are unremarkable, which is the saturation of Section 3.9.1 seen gene by gene.

3.9.5 Signal density across datasets

Table 11 places the two real analyses in context. Gene-level signal density varies by more than an order of magnitude between the maize MLM analysis and the human meta-analysis, and the dependence of the pathway statistic on pathway size tracks it closely, from −0.179 in the sparsest dataset to −0.807 in the densest. This is not a bias to be corrected away. In a saturated regime larger sets genuinely do accumulate more association signal, because the background itself

is enriched; the statistic reports something true about the data and uninteresting about the biology.

We note in passing that the maize GLM analysis, included here for contrast, shows a median gene-level p-value of 0.185, far below the 0.5 expected under the null. This is population-structure inflation rather than signal, and is why the MLM analysis is the one we report in Section 3.8. It also illustrates how readily a saturated-looking pathway result can arise from a mis-specified gene-level model rather than from biology, which is an argument for inspecting the gene-level null before interpreting any pathway output.

4 Discussion

4.1 What the Westfall–Young layer buys, and what it does not

The contribution of the Westfall–Young layer is narrow and should be stated narrowly. Reading the null matrix across its rows rather than down its columns costs nothing—the draws have already been taken—and it makes the family-wise statement exact under the actual overlap structure of the pathway database being used, without requiring that structure to be estimated or even described. It does not change any pathway's marginal evidence, which is what the column-wise reading gives, and it does not improve power. In the shared-universe simulations with no within-pathway correlation, where the analytic p-value is valid, the Westfall–Young threshold was 1.21 to 1.38 times the Bonferroni threshold 3.33×10^{-4}: modestly more permissive, which is the gain from accounting for overlap at an average of 8.8 pathways per gene. That is enough to matter for a borderline result and not enough to rescue an analysis that has gone wrong for other reasons.

The practical value is diagnostic as much as inferential. The ratio of the Bonferroni threshold to the realized Westfall–Young threshold is a single number that tells an analyst how far from independent their pathway family actually is. A ratio near one says the database behaves like a set of independent tests and Bonferroni is close to right; a ratio far from one says the correction in common use is substantially miscalibrated for this particular collection of gene sets. That number is cheap to compute as a by-product of the procedure and, as far as we are aware, is not routinely reported.

The ratio has to be read on calibrated statistics to mean this. In the same simulations with within-pathway correlation, the threshold set on the analytic scale was 5 to $5,117$ times *stricter* than Bonferroni—not because the pathway family was more dependent, but because the analytic p-value was anti-conservative and the Westfall–Young threshold, drawn from the same null, corrected for it. On that scale the ratio mixes the miscalibration of the per-test statistic with the dependence among tests and cannot separate them.

We were initially wrong about one aspect of this and correct the record here. We expected the size-standardized Westfall–Young variant of Section 2.3 to remove the dependence of pathway evidence on pathway size, on the reasoning that expressing each statistic against its own null bank puts all pathways on a common scale. It did not: in the maize data the size–evidence correlation moved from -0.807 to -0.800 in the human analysis and remained essentially unchanged elsewhere. The reason is visible once measured. The standard deviations of the per-size null banks differ by only a few percent across the pathway sizes we examined, so the standardization divides by a nearly constant factor and cannot remove a dependence that lives in the numerator. The size dependence is in the observed statistics, and in a saturated regime it is real accumulation of signal, not a scaling artefact. Standardization is still worth reporting because it changes individual rankings, but it is not a correction for size bias and should not be presented as one.

4.2 Saturation and the self-contained hypothesis

The LDL-cholesterol result in Section 3.9 is, we think, the most consequential finding in this paper, and it is not about CATFISH. Self-contained gene-set tests ask whether the genes in a set show more association than the null distribution of no association (Goeman and Bühlmann, 2007; Maciejewski, 2014). That question has a definite answer, and at $n \approx 1.3$ million for a highly polygenic trait the answer is yes for almost every set of genes, because 11.2% of all genes are individually genome-wide significant. A random set of fifty genes rejects the self-contained null with probability 0.997. Nothing has malfunctioned; the hypothesis being tested has simply stopped discriminating.

Three consequences follow. First, as GWAS sample sizes grow, self-contained testing will fail on more traits, and it will fail silently, producing long lists of highly significant pathways that look like results. Second, no amount of multiple-testing sophistication fixes this: we applied exact family-wise control and 1,374 of 1,390 pathways remained significant. Reporting a more careful threshold on a saturated analysis produces a more carefully wrong answer. Third, the competitive hypothesis—whether the genes in a set are *more* associated than genes outside it—does not saturate in the same way, because the comparison is against the observed genome-wide background rather than against no association. For large, well-powered human GWAS, the competitive framing is the appropriate one, and the self-contained framing that CATFISH implements is best suited to the low-signal, strongly polygenic regime illustrated by the maize analysis, where no gene reaches genome-wide significance and a ranking is the only available output.

This is a boundary on the applicability of the method we are presenting, and we would rather state it than discover it in review. It also explains an initially puzzling observation: the pathways CATFISH ranked highest in the human analysis were things like spliceosome assembly and the TCA cycle, which have no particular relationship to lipid metabolism but are large and contain many of the genes that are significant in a saturated genome.

4.3 Numerical precision is a correctness issue

Section 2.4 reads like an implementation note and is not one. Had we not moved to log space, this paper would have reported a top-pathway list that was determined by file order, with every leading p-value equal to the double-precision floor, and nothing in the output would have indicated that anything was wrong. We encountered exactly this failure twice during development, once from ACAT overflow and once from underflow of the combined p-value, and in both cases the symptom was a plausible-looking table.

The simulations produced a third instance, in a different component and with a different symptom. The soft TFisher implementation returns the null distribution function of its statistic, and the p-value is one minus that value; past $1 - 10^{-16}$ the difference is exactly zero. In the exchangeable design at $\rho = 0.6$, $m = 50$, about 3.5% of null pathways reached that point and were reported at the general lower floor. With the floor at 10^{-50} the empirical

family-wise error rate in that configuration was 0.24. Lowering the floor to 10^{-300}, in an attempt to recover resolution, raised it to 1.00: the spurious value became more extreme than anything another component produced, so the omnibus family minimum was the same shared number in essentially every simulated family and the Westfall–Young threshold sat on the tie. The fix was to cap the component at the precision it has. At 10^{-16}, the components with closed-form tails—which in the same configuration resolve genuinely distinct values below 10^{-100}—determine the family minimum, and it is once again a continuous statistic. A floor is a claim about resolution; a floor set below the resolution a computation actually has manufactures ties that look like extreme evidence.

We therefore suggest that any pathway method intended for use on modern large-n GWAS summary statistics should state explicitly how it represents p-values below 10^{-300}, and that reviewers should ask. The arithmetic is not difficult; the problem is that the failure is invisible unless specifically checked for. A simple diagnostic is to count how many pathways share the smallest reported p-value. If that count exceeds one or two, the ranking below that point is not meaningful. For a family-wise procedure the corresponding check is the number of distinct values taken by the null family minimum; a threshold drawn from a distribution with a large atom is not a threshold.

4.4 Practical recommendations

For users, our recommendations are as follows. Inspect the gene-level null before running any pathway analysis: a median gene-level p-value far from 0.5 indicates either genuine saturation or population-structure inflation, and these require different responses (the former argues for competitive testing, the latter for a mixed model). Calibrate against the correlated null whenever gene–gene correlations are available, and verify that the correlation file covers all chromosomes; in one of our own runs only chromosome 1 correlations had been parsed, and 85% of the genome was silently treated as independent, a failure that produced no error and no obviously wrong output. Report the per-pathway calibrated p-values and the Westfall–Young family-wise ones, together with the Bonferroni threshold and its ratio to the Westfall–Young threshold computed on calibrated statistics (Section 4.1), since the comparison is informative about the database. Before trusting any family-wise threshold, count the distinct values taken by the null family minimum. When using adaptive soft TFisher, choose a truncation grid that spans small and large thresholds; a grid confined to small τ leaves the statistic empty on most small pathways. Finally, in the low-signal regime, treat the ranking rather than the significance list as the deliverable, and validate it against independent biology where possible, as we have attempted to do with the maize analysis.

5 Limitations and future work

Several limitations bound what can be concluded from this work, and we list them explicitly.

No simulated GWAS. The simulations in Section 2.6 operate at the level of the omnibus statistic and its null, using block-structured and exchangeable correlation models for gene–gene dependence. We did not simulate genotypes, assign causal variants, generate phenotypes, and push them through the full GWAS–MAGMA–CATFISH pipeline. That experiment is the natural next step and is the one that would establish power against realistic genetic architectures rather than against latent mean shifts; it was deliberately excluded here rather than attempted partially. Consequently, the archetype power curves of Section 3.7 are power against stylised signal shapes injected on the gene-level z scale, not against realistic genetic architectures.

Unreported correlations are set to zero. The correlation matrices of Section 2.3 are built by filling MAGMA's reported gene pairs into an identity matrix, so any pair MAGMA does not report is treated as independent. For distant gene pairs this is close to true and the matrix is banded by construction, but a nearby pair that goes unreported—because it fell outside a batch boundary, or because one gene was dropped—becomes a genuine understatement of dependence, in the anti-conservative direction. Section 3.6 quantifies the cost, and corrects an assumption we made before measuring it: the two readings do *not* inherit the limitation equally. The family-wise reading degrades faster than the marginal one, because the family minimum is a tail statistic and an error in the correlation structure does its worst damage in the tail. Verifying chromosome coverage of the correlation file catches the gross form of this failure; `10_pair_coverage.R` measures the fine-grained form, and we recommend reporting it alongside any family-wise p-value.

Simplified LD models. Real gene–gene correlation is driven by physical proximity and decays continuously with distance. The shared-universe simulations approximate this with independent blocks of exchangeable correlation, so a pathway spanning several blocks has mostly uncorrelated members; the exchangeable design instead correlates every member of a pathway, which is far stronger than block-structured LD produces and is used only as a stress test. Neither reproduces the gradual decay of correlation with distance. The real-data analyses use empirical MAGMA correlations and are not subject to this limitation.

What strong control rests on. The argument of Section 2.3 for strong family-wise control is exact under its model and inherits that model's limits. It assumes association acts as a shift in the means of associated genes' z-scores, leaving the correlation among unassociated genes unchanged, and it defines a null pathway as one containing no associated gene. Under linkage disequilibrium a gene can be associated at the gene level solely because it is in LD with a causal variant in a neighbouring gene; a pathway containing it is then non-null under the self-contained gene-level hypothesis even if it contains no causal gene, so rejecting it is not a family-wise error, although it may be a false positive biologically. The argument also inherits the adequacy of the correlation matrix, and Section 3.6 shows how quickly family-wise calibration degrades when that matrix is incomplete. We have not separately confirmed strong control by simulation; under the model it follows from the complete-null results.

Parametric null. The null draws come from a multivariate normal model of gene–gene correlation rather than from permuting phenotypes. The procedure is Westfall–Young in construction, but its validity is that of the correlation model, which Section 3.6 makes concrete.

Soft TFisher resolution. With the current implementation the soft TFisher p-value cannot be resolved below about 10^{-16}, so among strongly enriched pathways that component ties. The other components resolve far beyond this and determine the omnibus ordering there; a log-scale implementation of the soft-thresholding tail would remove the limitation.

No competitive arm. We were unable to include a direct comparison against MAGMA's competitive gene-set test in this version, because the competitive entry point is not currently exported from the CATFISH package namespace. Given the argument of Section 4.2, that comparison is the single most important addition to make, and it is in progress.

Incomplete cross-species replication. The Westfall–Young analysis was run on maize and human data. The sorghum stem-volume dataset used in the original CATFISH paper could not be re-analysed under the new calibration because the MAGMA `.genes.raw` file, which the genome-wide correlated draw requires, was not available in the archived results; its signal-density summary in Table 11 is reported from the gene-level output alone.

Tail resolution. Monte Carlo calibration resolves p-values only to $1/(B+1)$. CATFISH offers a generalized Pareto tail extrapolation for the extreme tail (Knijnenburg et al., 2009), but it requires a minimum number of valid null draws, and a single missing draw can silently disable it; in one of our human runs this caused 638 of 1,348 pathways to tie at the Monte Carlo floor. The threshold is now set to half the number of permutations rather than to the full number, but we have not yet verified the fix at production scale.

Dependence on upstream choices. Results remain sensitive to the SNP-to-gene window, the reference panel used for LD, gene model definitions, and the pathway database, as is true of every method in this class. The 10 kb windows used here are conventional but not optimal for any particular trait.

6 Conclusion

Pathway analysis of GWAS summary statistics faces two distinct dependence problems, and the literature has concentrated on the easier one. Correlation among genes within a pathway is widely acknowledged and routinely modelled; correlation among pathways, induced by the shared gene membership that is intrinsic to how pathway databases are built, is usually handled with a correction that assumes it away. We have added a Westfall–Young max-statistic layer to CATFISH that controls the family-wise error rate in the strong sense whatever that dependence, under the same correlation model used for calibration, requires nothing to be estimated about it beyond the gene–gene correlation already used for calibration, and yields as a by-product a direct measure of how far Bonferroni is from correct for any given database, provided it is read on calibrated statistics.

Two further findings matter beyond this particular method. Computing the omnibus in log space is a correctness requirement at modern GWAS sample sizes, not an optimisation, and the failure it prevents is silent. And self-contained pathway testing saturates: in a GWAS of 1.3 million individuals, 11.2% of genes are genome-wide significant, a random fifty-gene set rejects the self-contained null with probability 0.997, and exact family-wise error control leaves 1,374 of 1,390 pathways significant. That is a property of the hypothesis, not of the correction. CATFISH, and self-contained pathway testing generally, is best suited to the low-signal polygenic regime, where the maize grain-nitrogen analysis recovered L-homoserine biosynthesis as the leading pathway and a known aspartate kinase isozyme as the leading gene out of 34,873, in agreement with independent prior work.

7 Conflicts of interest

The authors declare that they have no competing interests.

8 Funding

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9 Data availability

The GLGC LDL-cholesterol summary statistics are available from the Global Lipids Genetics Consortium. The CATFISH R package and all code used to generate the figures, tables and simulations underlying this article are available at github.com/nirwantandukar/CATFISH, in the `catfish2/` directory.

10 Author contributions statement

N.T. conceived and implemented the method, designed and ran the simulations, analysed the real data, and wrote the manuscript. J.-Y.T. advised on the statistical framework and calibration. R.R.-A. supervised the work and revised the manuscript. All authors read and approved the final manuscript.

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Table 4 Component-breakage stress test on overlapping pathways. Each of k component tests is miscalibrated by raising its p-value to the power γ; $\gamma < 1$ makes a component conservative and $\gamma > 1$ anti-conservative. Marginal columns are per-pathway type I error; FWER columns are family-wise error rates over the K pathways tested, measured on held-out draws. Nominal level $\alpha = 0.05$.

LD	ρ	m	Direction	γ	k	Marginal		FWER	
						Analytic	MVN	WY	Bonf.
independent	0.0	20	anticonservative	1.4	0	0.0631	0.0489	0.0640	0.0560
independent	0.0	20	anticonservative	1.4	1	0.0847	0.0491	0.0620	0.1313
independent	0.0	20	anticonservative	1.4	2	0.1013	0.0488	0.0653	0.1500
independent	0.0	20	anticonservative	1.4	3	0.1195	0.0490	0.0620	0.2373
independent	0.0	20	anticonservative	1.4	4	0.1372	0.0489	0.0633	0.2967
independent	0.0	50	anticonservative	1.4	0	0.0682	0.0517	0.0700	0.0560
independent	0.0	50	anticonservative	1.4	1	0.0917	0.0520	0.0553	0.1347
independent	0.0	50	anticonservative	1.4	2	0.1110	0.0515	0.0600	0.1453
independent	0.0	50	anticonservative	1.4	3	0.1316	0.0516	0.0627	0.2247
independent	0.0	50	anticonservative	1.4	4	0.1502	0.0512	0.0660	0.2840
moderate	0.3	20	anticonservative	1.4	0	0.0848	0.0492	0.0440	0.1260
moderate	0.3	20	anticonservative	1.4	1	0.1100	0.0489	0.0453	0.2227
moderate	0.3	20	anticonservative	1.4	2	0.1251	0.0492	0.0487	0.2320
moderate	0.3	20	anticonservative	1.4	3	0.1442	0.0496	0.0520	0.3147
moderate	0.3	20	anticonservative	1.4	4	0.1624	0.0496	0.0487	0.3627
moderate	0.3	50	anticonservative	1.4	0	0.1228	0.0467	0.0380	0.1700
moderate	0.3	50	anticonservative	1.4	1	0.1515	0.0465	0.0420	0.2627
moderate	0.3	50	anticonservative	1.4	2	0.1675	0.0463	0.0420	0.2707
moderate	0.3	50	anticonservative	1.4	3	0.1891	0.0463	0.0460	0.3447
moderate	0.3	50	anticonservative	1.4	4	0.2095	0.0463	0.0407	0.3833
strong	0.6	20	anticonservative	1.4	0	0.1142	0.0604	0.0780	0.1740
strong	0.6	20	anticonservative	1.4	1	0.1404	0.0607	0.0787	0.2527
strong	0.6	20	anticonservative	1.4	2	0.1530	0.0606	0.0787	0.2613
strong	0.6	20	anticonservative	1.4	3	0.1723	0.0612	0.0727	0.3153
strong	0.6	20	anticonservative	1.4	4	0.1898	0.0607	0.0747	0.3427
strong	0.6	50	anticonservative	1.4	0	0.1761	0.0435	0.0420	0.2620
strong	0.6	50	anticonservative	1.4	1	0.2017	0.0437	0.0447	0.3140
strong	0.6	50	anticonservative	1.4	2	0.2115	0.0438	0.0447	0.3153
strong	0.6	50	anticonservative	1.4	3	0.2290	0.0425	0.0380	0.3547
strong	0.6	50	anticonservative	1.4	4	0.2474	0.0432	0.0387	0.3887
independent	0.0	20	conservative	0.7	0	0.0631	0.0489	0.0640	0.0560
independent	0.0	20	conservative	0.7	1	0.0554	0.0486	0.0627	0.0493
independent	0.0	20	conservative	0.7	2	0.0479	0.0488	0.0587	0.0480
independent	0.0	20	conservative	0.7	3	0.0380	0.0488	0.0687	0.0353
independent	0.0	20	conservative	0.7	4	0.0266	0.0485	0.0647	0.0207
independent	0.0	50	conservative	0.7	0	0.0682	0.0517	0.0700	0.0560
independent	0.0	50	conservative	0.7	1	0.0601	0.0516	0.0700	0.0473

Table 5 Leave-one-out component sensitivity on overlapping pathways. Each row removes a single component and rebuilds the null bank, the marginal calibration and the Westfall–Young threshold from scratch. Nominal level $\alpha = 0.05$.

LD	ρ	m	Dropped	Marginal		FWER	
				Analytic	MVN	WY	Bonf.
independent	0.0	20	none	0.0631	0.0489	0.0640	0.0560
independent	0.0	20	ACAT	0.0647	0.0492	0.0640	0.0620
independent	0.0	20	Fisher	0.0648	0.0488	0.0560	0.0540
independent	0.0	20	Stouffer	0.0630	0.0486	0.0700	0.0480
independent	0.0	20	minP	0.0649	0.0487	0.0660	0.0620
independent	0.0	20	TFisher	0.0554	0.0489	0.0660	0.0440
independent	0.0	50	none	0.0682	0.0517	0.0700	0.0560
independent	0.0	50	ACAT	0.0701	0.0519	0.0700	0.0660
independent	0.0	50	Fisher	0.0717	0.0517	0.0720	0.0500
independent	0.0	50	Stouffer	0.0703	0.0514	0.0680	0.0540
independent	0.0	50	minP	0.0705	0.0518	0.0700	0.0660
independent	0.0	50	TFisher	0.0589	0.0515	0.0680	0.0360
moderate	0.3	20	none	0.0848	0.0492	0.0440	0.1260
moderate	0.3	20	ACAT	0.0885	0.0495	0.0440	0.1420
moderate	0.3	20	Fisher	0.0851	0.0489	0.0440	0.1160
moderate	0.3	20	Stouffer	0.0789	0.0504	0.0580	0.0980
moderate	0.3	20	minP	0.0895	0.0495	0.0440	0.1420
moderate	0.3	20	TFisher	0.0780	0.0491	0.0420	0.1140
moderate	0.3	50	none	0.1228	0.0467	0.0380	0.1700
moderate	0.3	50	ACAT	0.1285	0.0467	0.0380	0.1900
moderate	0.3	50	Fisher	0.1229	0.0466	0.0400	0.1680
moderate	0.3	50	Stouffer	0.1098	0.0471	0.0500	0.1680
moderate	0.3	50	minP	0.1293	0.0467	0.0380	0.1900
moderate	0.3	50	TFisher	0.1131	0.0456	0.0340	0.1460
strong	0.6	20	none	0.1142	0.0604	0.0780	0.1740
strong	0.6	20	ACAT	0.1196	0.0610	0.0780	0.1820
strong	0.6	20	Fisher	0.1132	0.0597	0.0800	0.1720
strong	0.6	20	Stouffer	0.1018	0.0625	0.0680	0.1600
strong	0.6	20	minP	0.1210	0.0612	0.0780	0.1820
strong	0.6	20	TFisher	0.1064	0.0603	0.0800	0.1660

Table 6 Per-component calibration. Each component test is calibrated against its own null rather than being combined into the omnibus, at $m = 50$. Analytic columns assume gene independence; calibrated columns use the correlated null. Nominal $\alpha = 0.05$.

LD	Component	Analytic	Calibrated	WY	λ_{an}	λ_{cal}
independent	ACAT	0.0555	0.0529	0.0700	1.008	1.014
independent	Fisher	0.0513	0.0513	0.0580	1.004	0.995
independent	Stouffer	0.0511	0.0499	0.0440	1.002	1.005
independent	minP	0.0545	0.0528	0.0700	1.014	1.014
independent	TFisher	0.0984	0.0516	0.0680	2.019	1.005
independent	Omnibus	0.0682	0.0517	0.0700	1.039	0.995
moderate	ACAT	0.0561	0.0521	0.0520	1.047	1.043
moderate	Fisher	0.1145	0.0455	0.0500	0.933	1.091
moderate	Stouffer	0.1293	0.0456	0.0380	1.015	1.062
moderate	minP	0.0517	0.0527	0.0520	0.868	1.005
moderate	TFisher	0.1401	0.0481	0.0500	1.943	1.043
moderate	Omnibus	0.1228	0.0467	0.0380	0.971	1.052
strong	ACAT	0.0555	0.0414	0.0280	1.048	0.959
strong	Fisher	0.1632	0.0411	0.0380	0.695	0.855
strong	Stouffer	0.1887	0.0501	0.0540	0.992	0.941
strong	minP	0.0431	0.0415	0.0280	0.485	0.798
strong	TFisher	0.1750	0.0391	0.0340	1.256	0.806
strong	Omnibus	0.1761	0.0435	0.0420	0.665	0.880

Table 7 Robustness to an incomplete gene–gene correlation file. Data are generated from the true correlation structure; calibration is built from a matrix in which the stated fraction of off-diagonal entries has been zeroed, as an unreported gene pair would be. Marginal columns are per-pathway type I error, FWER columns are family-wise rates over K pathways. Nominal $\alpha = 0.05$.

LD	Dropped	Marginal		FWER		λ	
		Analytic	MVN	WY	Bonf.	Analytic	MVN
moderate	0%	0.1300	0.0535	0.0525	0.1975	0.857	1.029
moderate	25%	0.1300	0.0642	0.0825	0.1975	0.857	0.883
moderate	50%	0.1300	0.0823	0.1225	0.1975	0.857	0.915
moderate	75%	0.1300	0.0922	0.1625	0.1975	0.857	0.831
moderate	100%	0.1300	0.1090	0.2075	0.1975	0.857	0.841
strong	0%	0.1584	0.0504	0.0525	0.2150	0.609	0.966
strong	25%	0.1584	0.0829	0.1025	0.2150	0.609	0.791
strong	50%	0.1584	0.1055	0.1375	0.2150	0.609	0.724
strong	75%	0.1584	0.1213	0.1675	0.2150	0.609	0.678
strong	100%	0.1584	0.1424	0.2200	0.2150	0.609	0.609

Table 8 Archetype verification under the family-wise decision rule. For each archetype and pathway size, the component achieving the highest family-wise power most often across effect scales, against the set predicted in Table 1. Cells in which no component or every component fires are excluded as uninformative. Worst regret is the largest gap in a single cell between the best component and the omnibus.

Archetype	m	Predicted	Observed winner	In set	Worst regret
SSS	5	ACAT, minP	minP	✓	0.108
SSS	20	ACAT, minP	ACAT	✓	0.131
SSS	50	ACAT, minP	minP	✓	0.039
MDS	5	Fisher, Stouffer, TFisher	Fisher	✓	0.050
MDS	20	Fisher, Stouffer, TFisher	Fisher	✓	0.012
MDS	50	Fisher, Stouffer, TFisher	Fisher	✓	0.029
WDE	5	Stouffer, Fisher	Fisher	✓	0.032
WDE	20	Stouffer, Fisher	Stouffer	✓	0.020
WDE	50	Stouffer, Fisher	Stouffer	✓	0.021
MSA	5	TFisher, Fisher, ACAT	Fisher	✓	0.042
MSA	20	TFisher, Fisher, ACAT	Fisher	✓	0.009
MSA	50	TFisher, Fisher, ACAT	Fisher	✓	0.021
SGP	5	minP, ACAT	ACAT	✓	0.137
SGP	20	minP, ACAT	minP	✓	0.355
SGP	50	minP, ACAT	minP	✓	0.476

Table 9 Power regret of the ACAT-O omnibus by archetype. Regret is the difference between the best single component and the omnibus in the same cell, over all pathway sizes and effect scales. The final two columns give the fraction of cells in which the omnibus is within 0.05 and 0.10 of the best component.

Code	Archetype	Mean	Median	Worst	≤ 0.05	≤ 0.10
MSA	Mixed Signal Architecture	0.006	0.001	0.042	100%	100%
MDS	Moderate Dense Signal	0.010	0.003	0.050	94%	100%
WDE	Weak Diffuse Enrichment	0.010	0.010	0.032	100%	100%
SSS	Strong Sparse Signal	0.025	0.003	0.131	83%	89%
SGP	Single-Gene Proxy	0.066	0.005	0.476	72%	83%

Table 10 Top five prioritized genes for LDL cholesterol. Rank columns are the gene's position within each evidence layer: locus-level GWAS (18563 genes with an overlapping SNP), MAGMA gene-level association (18563 genes), and CATFISH pathway enrichment (9727 pathway-annotated genes). Score is the composite of Section 2.5. The full top fifty are in Supplementary Table S1.

Rank	Gene	Score	GWAS	MAGMA	Pathway	Leading pathway
1	*LDLR*	8.11	6	4	462	plasma lipoprotein assembly remod…
2	*PVRL2*	7.73	4	3	4298	immunoregulatory interactions bet…
3	*PCSK9*	7.53	10	8	462	plasma lipoprotein assembly remod…
4	*APOE*	6.80	2	308	462	plasma lipoprotein assembly remod…
5	*SMARCA4*	6.76	32	29	256	atp dependent chromatin remodelers

Table 11 Gene-level signal density and pathway-size dependence across the four GWAS analysed. The last column is the Spearman correlation between pathway size and the pathway-level evidence ($-\log_{10}$ omnibus p); more negative values indicate that larger pathways receive systematically stronger evidence. n.c., not computed.

Dataset	Genes GW significant	Median gene p	Size correlation
Maize grain N (MLM)	0.00%	0.468	-0.179
Sorghum stem volume	0.86%	0.440	n.c.
Maize grain N (GLM)	1.15%	0.185	-0.667
Human LDL-C (GLGC, EUR)	11.2%	0.083	-0.807

Supplementary material

Table 12 Top fifty prioritized genes for LDL cholesterol, with per-layer ranks. Layer sizes as in Table 10 (18563 / 18563 / 9727 genes).

Rank	Gene	Score	GWAS	MAGMA	Pathway	Leading pathway
1	*LDLR*	8.11	6	4	462	plasma lipoprotein assembly remod...
2	*PVRL2*	7.73	4	3	4298	immunoregulatory interactions bet...
3	*PCSK9*	7.53	10	8	462	plasma lipoprotein assembly remod...
4	*APOE*	6.80	2	308	462	plasma lipoprotein assembly remod...
5	*SMARCA4*	6.76	32	29	256	atp dependent chromatin remodelers
6	*APOC1*	6.65	2	302	462	plasma lipoprotein assembly remod...
7	*LRRC16A*	6.57	262	1	1876	factors involved in megakaryocyte...
8	*KIAA1324*	6.54	50	2	–	–
9	*TOMM40*	6.25	2	248	4722	protein localization
10	*ABCG5*	6.23	28	28	2180	disorders of transmembrane transp...
11	*LPA*	6.20	40	35	462	plasma lipoprotein assembly remod...
12	*APOB*	6.16	15	185	462	plasma lipoprotein assembly remod...
13	*ERCC2*	6.14	70	21	1803	rna polymerase i transcription
14	*GPAM*	6.08	278	9	530	estrogen dependent gene expression
15	*RELB*	6.04	16	68	2484	somatic hypermutation of immunogl...
16	*DOCK6*	6.01	192	6	1876	factors involved in megakaryocyte...
17	*LDLRAP1*	5.97	251	14	462	plasma lipoprotein assembly remod...
18	*DNM2*	5.93	44	47	2065	clathrin mediated endocytosis
19	*EHMT2*	5.89	759	13	96	cellular senescence
20	*ABCG8*	5.87	28	64	2180	disorders of transmembrane transp...
21	*SPC24*	5.85	33	39	4078	mitotic prometaphase
22	*IGF2R*	5.78	138	16	2065	clathrin mediated endocytosis
23	*ABCA1*	5.78	273	18	462	plasma lipoprotein assembly remod...
24	*MAP2K6*	5.75	119	128	96	cellular senescence
25	*ANKDD1B*	5.70	36	19	–	–
26	*APOC2*	5.68	12	626	462	plasma lipoprotein assembly remod...
27	*CETP*	5.58	120	59	462	plasma lipoprotein assembly remod...
28	*HLA-DQA1*	5.57	223	26	862	interferon gamma signaling
29	*CBLC*	5.56	9	106	–	–
30	*BSND*	5.44	63	37	5394	ion channel transport
31	*GATAD2A*	5.41	53	389	256	atp dependent chromatin remodelers
32	*HIST1H2AC*	5.40	184	271	96	cellular senescence
33	*CEACAM16*	5.39	18	76	–	–
34	*PPP1R13L*	5.37	70	48	3710	regulation of tp53 activity
35	*HMGCR*	5.35	22	356	1304	metabolism of steroids
36	*USP24*	5.35	20	165	3206	ub specific processing proteases
37	*CELSR2*	5.33	8	213	–	–
38	*KANK2*	5.31	31	54	–	–
39	*HIST1H4C*	5.31	174	365	96	cellular senescence
40	*POLK*	5.31	35	195	2780	homology directed repair
41	*APOC4*	5.30	12	1174	462	plasma lipoprotein assembly remod...
42	*PHC1*	5.30	549	72	96	cellular senescence
43	*NPC1L1*	5.26	127	20	8228	digestion and absorption
44	*BCAM*	5.26	5	380	–	–
45	*GSTM2*	5.20	252	30	1581	phase ii conjugation of compounds

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